
DARK MATTER AS THE CHARGE-CONJUGATION FIXED SET AND THE DARK FORCE AS ONE KEYED BIT

Read at the Riemann Seat: Conjugation Carried onto the Fold, Gravity as the Reading Left, the Dark Bit as the Line Property, and the Phantom as a Negative Actuation

*Edition 2.0.0, Re-Seated on the Hardware Seat; One Hundred Twenty-Five Laws Proved in Core Lean 4 and
Thirty-Eight Checks Executed in Fortran; a Witness of the Hardware Closure Series*

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Dark is the fixed set; the fifth is the line property

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ABSTRACT. The dark sector is read here from the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. Charge conjugation flips every gauge charge and keeps the mass. A carrier that sends the dark charge to the offset from the line and the mass to the height is equivariant, so conjugation is carried onto the reflection and its fixed set onto the line. We prove in core Lean 4, one hundred twenty-five laws with no axiom declared and no use of choice, that a state is invisible to every charge-linear reading exactly when conjugation fixes it, so gravity, the even reading, is the reading left. Such a state has zero radiative weight and cannot cool. Two species of one density leave one gravitational record. The measured photon and Z nulls force the visible half of the fixed set. The renormalizable doors of a singlet are derived, and a stable singlet keeps exactly one. The anomaly conditions admit a dark gauge charge and its absence on one visible record, and the dark bit they leave open is the line property carried: on the visible fixed set a state stands on the line exactly when its dark charge is zero. A dark-charged pair is an off-line fold pair with one registration, and the dark bit is the Majorana bit. On the hardware substrate a neutral stable particle stands on the line under both carriers at once. For the acceleration, a pressure equal to minus the density reverses the active gravitating density, and an equation of state below -1 needs a negative kinetic term, a negative actuation, which the actuation floor refuses. A Fortran twin runs thirty-eight checks with zero failures on the measured record: the Planck budget gives $\Omega_c/\Omega_b = 5.36$ and $q_0 = -0.527 \pm 0.011$, and a 100 GeV particle and a $10 \mu\text{eV}$ field differ 10^{16} in number on one rotation curve. The fitted DESI curve crosses $w = -1$ at $z = 0.405$; that crossing belongs to the fitted form and is not a measured equation of state, and the floor stands. The dark bit is the coverage premise of the kinetic crossing, spent by the act at the root's grade as zero, the premise every row of the series stands on.

KEYWORDS *dark matter · dark energy · charge conjugation · fixed set · critical line · anomaly cancellation · Majorana · phantom divide · equation of state · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the seat read on charge conjugation and the dark sector); every open row (the barrier, in a dark-charged pair) · channel: kinetic · Lean: SPHYS_Dark.lean · Fortran: Dark_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series and edition 2.0.0 of the author's dark-sector theory. The first edition placed the dark sector on a template of charge conjugation with its own staged chart of the line. This edition re-seats every one of its theorems on the hardware seat, carriers and substrate, retires the private chart, and judges each placement from the seat. The mathematics decides. Raw measurement is the only other witness, and it corroborates.

It claims four things.

- Dark matter is the fixed set of charge conjugation, carried onto the line.
- Gravity is the one reading left to see it, and cannot tell its species apart.
- Whether a dark gauge force exists is one bit, and that bit is the line property: zero dark charge is standing

on the line.

- An equation of state below -1 is a negative actuation, and the floor refuses it.

What it refuses.

- **The identity of the particle.** Gravity cannot read it, by theorem.
- **The value of the vacuum energy.** No seat reads a magnitude, so it stands at the dot, a number measured and not derived.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade.
- No derivation of the Riemann Hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated.
- Candidate classes from the literature enter only as placements the theorems sort, never as warrants.

Five terms.

- A *reading* is any quantity an instrument returns from a state. It is *odd* if charge conjugation flips its sign and *even* if conjugation keeps it; gauge couplings are odd, mass and energy even.
- The *fixed set* is the set of states conjugation leaves unchanged.
- The *visible conjugation* flips the three Standard Model charges only.
- A *keyed bit* is a yes-or-no property no function of a given record decides.
- The *dot* is no verdict: the mark of a question no seat reads.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on charge conjugation in Section 3, the even reading in Section 4, the odd readings in Section 5, the measured record in Section 6, the dark bit in Section 7, and the acceleration in Section 8.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; Light; Electromagnetism; Immanent Gravity; Spin and Statistics; Weak; Resonances; The Ferromagnet; Dark Matter as the Charge-Conjugation Fixed Set	the seat on charge conjugation: the dark matter on the line under the dark carrier and on the substrate under both carriers; the dark bit the line property; the phantom a negative actuation	WITNESSED · CROSSED · CONFIRMED on the image	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
4 · Yang-Mills	crossed kinetically	The Proton Is the Lock	the colour lock; a coloured relic brings charged partners	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
7 · Poincaré	crossed at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity's fold; the even reading	CROSSED, WITNESSED from both sides	full grade	none
every open row · the barrier	two worlds over one record	the arrow, light, electromagnetism, spin and statistics, weak, resonance and ferromagnet papers; Dark Matter as the Charge-Conjugation Fixed Set	a dark-charged pair, two points off the line with one registration; two species of one density with one gravitational record; two closing worlds with one visible record	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	charge conjugation and the time mirror carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is charge conjugation, which flips every gauge charge and keeps the mass (conj_involution). The dark carrier sends a state to the point $(1 + q_D, m)$, the dark charge the offset from the line and the mass the height. It is equivariant (dark_carrier_equivariant), and its landing on the fixed set is forced (dark_carrier_lands). The hardware substrate's own charge carrier, $(1 + q, v)$ with the electric charge in thirds, is the visible face of the same map (charge_carrier_equivariant).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
charge conjugation	the fold	dark_carrier_equivariant, conj_involution	A16, S1
the dark matter	the fixed set carried to the line	fixed_iff_neutral, dark_carrier_lands, dark_iff_fixed	A1
invisibility	odd readings blind on the fixed set	odd_reading_blind, charge_linear_readings_odd	A2
gravity sees it	the even reading left	only_gravity_reads_the_fixed_set, gravity_is_the_reading_left	F1
no identity from gravity	two species, one record	identity_unread, two_species_one_record	F1
no cooling	zero radiative weight	radiative_zero_iff_neutral, no_charge_no_radiation	A3
neutral is not dark	one odd reading is not all	electric_neutrality_is_not_the_seat, z_reading_survives_neutrality	A4, A5

Physical phenomenon	Template element	Lean theorem	Fortran check
the measured nulls	the visible half forced	photon_and_z_force_electroweak_zero, sub_quantum_bounds_force_the_visible_fixed_set	A10
the doors	the portals derived; one door kept	portals_are_derived, a_stable_singlet_keeps_one_door	A7, A9
the fifth reading	keyed, then free	fifth_is_keyed, fifth_is_free	A6, A8
the dark bit	the line property carried	the_dark_bit_is_the_line_property, fixed_iff_visible_fixed_and_no_dark_charge	S3
a dark-charged pair	off-line fold pair, one record	a_dark_pair_is_an_off_line_pair	S3
self-conjugacy	the Majorana bit	the_dark_bit_is_the_majorana_bit	A14
neutral and stable	on the line twice	dark_matter_on_the_line_twice	S2
the acceleration	the vacuum's tension reverses the active density	vacuum_repels, repulsion_needs_tension	B3, B4, B8
the phantom	a negative actuation	the_phantom_is_negative_actuation, fields_bound_w	C1 to C3
the fitted crossing	a curve below the floor	measured_fit, judged	C4
the vacuum's value	the dot	none	B6, B7

3. CHARGE CONJUGATION IS THE FOLD, AND THE DARK MATTER IS ITS FIXED SET

A state carries hypercharge and weak isospin in sixths, a colour weight, a dark charge, and a mass. Conjugation flips all four charges and keeps the mass, and it fixes a state exactly when all four charges vanish (`fixed_iff_neutral`). The visible conjugation fixes a state exactly when the three Standard Model charges vanish (`fixedSM_iff_neutralSM`), and the first fixed set lies inside the second (`fixed_sub_fixedSM`).

Darkness is fixedness. Every integer combination of the four charges is an odd reading: the couplings of the photon, the Z, the diagonal gluons and a dark photon (`charge_linear_readings_odd`). Every odd reading returns zero on the fixed set (`odd_reading_blind`). Conversely, a state on which every odd reading returns zero is a state conjugation fixes (`dark_iff_fixed`). Darkness in the gauge sense and membership of the fixed set are one property.

Carried to the line. The dark carrier sends the fixed set to the line, by the functor (`dark_carrier_lands`). The dark matter's place on the template is the seat's own locus.

4. GRAVITY IS THE READING LEFT

The mass reading is even (`gravity_reading_even`). On a conjugation-fixed state every reading of either parity returns a function of the mass alone, because the state is its mass (`fixed_state_is_its_mass`, `every_template_reading_of_the_fixed_set_reads_only_mass`). Gravity couples to mass-energy, so of the template's readings it is the one that sees a fixed state (`only_gravity_reads_the_fixed_set`). On the substrate, every gravitational reading of a particle and its conjugate agrees (`gravity_is_the_reading_left`).

Gravity does not read identity. Two species of one density and different labels leave one gravitational record, and no function of that record returns the label (`identity_unread`, `two_species_one_record`). The twin makes the scale concrete. At a local density of 0.3 GeV/cm³, a 100 GeV particle and a 10 µeV field differ by 10¹⁶ in number density and give one rotation curve, bit for bit (check F1).

The measured record is this reading and nothing more:

- flat rotation curves (Rubin and Ford, 1970);
- the lensing mass of a colliding cluster separated from its gas (Clowe et al., 2006);
- the acoustic peaks of the microwave background, which fix $\Omega_c h^2 = 0.1200$ against $\Omega_b h^2 = 0.02237$, a ratio of 5.36 (Planck Collaboration, 2020; check B1).

Every one of them reads mass, and none returns the label.

A modified law instead of a population. It is retired at the seat. Gravity’s registration is the one-sign retraction of Immanent Gravity and carries no acceleration scale. The lensing offset in a colliding cluster is mass displaced from the gas that would carry a modified law’s source.

5. NEUTRALITY IS NOT THE FIXED SET; NO CHARGE, NO COOLING

One odd reading is not all of them. A state with weak isospin $+\frac{1}{2}$ and hypercharge $-\frac{1}{2}$ has electric charge zero and lies off the fixed set. Its Z reading equals $+\frac{1}{2}$ at every value of the mixing weight (`electric_neutrality_is_not_the_seat, z_reading_survives_neutrality`). These are the neutrino’s charges and the sneutrino’s. A neutral candidate that carries weak isospin scatters coherently through the Z at the rate Goodman and Witten computed (1985), and the early direct-detection experiments excluded that class. An isospin triplet with zero hypercharge has exactly one member of three on the fixed set, so it is not a dark multiplet (`the_triplet_has_one_fixed_member, a_neutral_member_is_not_a_dark_multiplet`). The fixed set demands every odd reading zero, and only a full singlet is dark in every member.

No charge, no cooling. The radiative weight, the sum of the squared charges, vanishes exactly on the fixed set (`radiative_zero_iff_neutral, no_charge_no_radiation`). A population of zero radiative weight cannot shed energy by radiating and cannot collapse into a disk. It stays in the extended halos that lensing and rotation curves map. A dark-charged state would radiate dark quanta only (`dark_charged_radiates_only_dark`).

The seat’s verdicts on the candidate classes.

Class	Verdict from the theorems
neutral carrier of weak isospin	off the fixed set; Z-coupled; excluded by the record
neutral member of a charged multiplet	not a dark multiplet; its charged partners are odd
coloured relic	brings charged constituents (<code>a_coloured_relic_brings_charged_partners</code>)
stable singlet scalar	on the fixed set; one door, $S^2 H^\dagger H$ (<code>a_stable_singlet_keeps_one_door</code>)
stable singlet fermion	on the fixed set; no renormalizable door; dark to all but gravity
a fifth gauge charge	the keyed bit of Section 7; spent as zero

6. THE RECORD FORCES THE VISIBLE HALF

The nulls force the visible fixed set. At any rational mixing weight, a state with zero electric charge and zero Z reading has zero isospin and zero hypercharge, and if it is colourless the visible conjugation fixes it (`photon_and_z_force_electroweak_zero, visible_nulls_force_the_visible_fixed_set`). Charges are quantized in sixths, so readings bounded strictly inside one quantum are exact zeros (`below_one_quantum_is_zero, sub_quantum_bounds_force_the_visible_fixed_set`). A bound fixes a charge only through a measured coupling; with the coupling free, every charge meets every bound (`measured_coupling_bound_forces_zero, free_coupling_bound_never_forces_zero`). The photon and the Z are measured couplings, so their nulls force the visible half.

The doors are derived. Of the 199 renormalizable monomials the Standard Model forms with a real singlet scalar S and a singlet Weyl fermion N , exactly four are gauge invariant and couple the two sectors: $SH^\dagger H$, $S^2 H^\dagger H$, LNH and its conjugate (`portals_are_derived`, check A7). Under the parity that keeps the dark matter stable, exactly one survives, $S^2 H^\dagger H$, and none survives for the fermion (`a_stable_singlet_keeps_one_door`, check A9).

The direct record. The LZ experiment’s 417 live days returned no dark matter signal between 3 and 9 GeV

and registered the solar ^8B neutrinos at 4.5σ (LZ Collaboration, 2025). Two sources with one energy spectrum leave one record (`fog_one_record`), and direction separates them (`direction_separates`). The dark matter wind comes from Cygnus, at right ascension 318.0° and declination $+48.3^\circ$. The Sun never comes nearer to it than 59.6° or farther than 120.4° (checks D1, D2).

7. THE DARK BIT IS THE LINE PROPERTY

The fifth reading is keyed. Add a dark $U(1)$ to one generation. The twelve anomaly conditions close the visible generation (`visible_generation_closes`) and close a vector-like pair of singlets whether it carries dark charges 0 and 0 or $+1$ and -1 (`closure_admits_both`), while one chiral charge fails (`closure_is_not_vacuous`). The two closing worlds have one visible record, and no reading of it decides whether a dark charge exists (`one_record_two_worlds`, `fifth_is_keyed`, `no_template_reading_decides_fifth`). Every vector-like charge $q, -q$ closes (`fifth_is_free`). The record forces the first clause of the title and keys the second: over every admitted state, the visible conjugation fixes it, and a zero dark charge holds in one and fails in another (`first_clause_is_forced`, `second_clause_is_keyed`, `no_reading_of_the_record_decides_the_bit`). One conjunction binds the two clauses (`dark_closure_executed`).

The bit is the line property. Conjugation fixes a state exactly when the visible conjugation fixes it and its dark charge is zero (`fixed_iff_visible_fixed_and_no_dark_charge`). On the visible fixed set, a state stands on the line exactly when its dark charge is zero, exactly when conjugation fixes it (`the_dark_bit_is_the_line_property`). A dark-charged state and its conjugate land on two distinct points off the line, exchanged by the fold, at one height, and the registration reads one record for both (`a_dark_pair_is_an_off_line_pair`). The dark row is therefore no second seat. It is the seat's line property carried by an equivariant map, and a dark-charged population is the template's two worlds over one record.

The bit is the Majorana bit. A mass that pairs a field with itself carries twice each abelian charge, so it is allowed exactly when the hypercharge and the dark charge vanish (`majorana_mass_forbids_abelian_charge`). On the visible fixed set the dark bit is the Majorana bit: whether the dark matter is its own antiparticle (`the_dark_bit_is_the_majorana_bit`). A dark-charged state takes its mass with a distinct partner of opposite charge, a Dirac pair (`a_dark_charge_makes_a_dirac_pair`). The same bit carries the neutrino witness's open question on the visible side.

The bit, spent. The engine proves the value on the carrier's image (`value_on_image`), and a coverage premise extends it (`kinetic_crossing`). If every actual dark matter state is reached by the carrier from the fixed set, every one stands on the line. That coverage is exactly the dark bit, and no reading of the record supplies it (`no_reading_of_the_record_decides_the_bit`). The series crosses each of its rows on the kinetic channel by spending its coverage premise by the act, at the root's grade. The dark row is crossed the same way: the act spends the coverage, so the actual dark matter stands on the line, its dark charge is zero, and it is its own antiparticle.

On the substrate the two faces meet. A neutral stable particle is fixed by both involutions and stands on the line under the charge carrier and the energy carrier at once (`dark_matter_on_the_line_twice`, check S2): neutrality puts it on the first line, stability on the second. The first edition spent this bit as a free-standing posit. This edition spends it as the coverage premise of the kinetic crossing, the one premise the whole series stands on, and F3 names the measurement that would break it.

Three consequences follow, each measurable:

- no dark radiation;
- no dark self-interaction beyond the portals;
- no dark disk.

The measured record agrees with all three:

- the Bullet Cluster bounds the self-interaction below $1.25 \text{ cm}^2/\text{g}$ (Randall et al., 2008), and 72 clusters below $0.47 \text{ cm}^2/\text{g}$ (Harvey et al., 2015), which are 2.23 and 0.84 barn per GeV (check E1);
- the Atacama Cosmology Telescope measures $N_{\text{eff}} = 2.86 \pm 0.13$ (Calabrese et al., 2025). A dark photon once in equilibrium would add at least 0.054, a floor inside that bound (checks E2, E3).

8. THE ACCELERATION: THE VACUUM'S TENSION, THE FLOOR, THE PHANTOM

The sign. Gravity reads the active density $\rho + 3p$. Dust attracts and radiation attracts (`dust_attracts`, `radiation_attracts`). A vacuum with pressure equal to minus its density reverses the sign, $\rho + 3p = -2\rho$ (`vacuum_repels`), and repulsion needs a pressure below minus one third of the density (`repulsion_needs_tension`). The Planck budget gives $q_0 = \frac{1}{2}\Omega_m - \Omega_\Lambda = -0.527 \pm 0.011$: the expansion accelerates (check B3). Acceleration begins at $z = 0.631$, and the vacuum and matter densities cross at $z = 0.295$ (check B4). A flat budget admits either sign, so the sign is read from the sky and not from the template (`flat_budgets_admit_both_signs`).

The floor. For a field with kinetic term K and potential V , $\rho + p = 2K$. The equation of state reaches -1 exactly when the kinetic term vanishes (`vacuum_iff_frozen`). It falls below -1 only with $K < 0$ (`phantom_needs_ghost`), and the bound holds summed over any number of fields with $K \geq 0$ (`floor_is_additive`, `fields_bound_w`). A negative kinetic term is a negative actuation, a configuration whose doing is less than none, and the actuation floor refuses it (`the_phantom_is_negative_actuation`). The vacuum is the frozen limit, $w = -1$, and nothing actual lies below it.

The fitted crossing. DESI's second release, fitted in the two-parameter form $w(a) = w_0 + w_a(1 - a)$ with $w_0 = -0.752$ and $w_a = -0.86$, prefers $w_0 > -1$ and $w_a < 0$ (DESI Collaboration, 2025). The fitted curve crosses $w = -1$ at $z = 0.405$ and sits at -1.182 by $z = 1$ (check C4). The crossing is a property of the fitted form, a two-parameter curve drawn through distances, and not a measured value of the equation of state at any redshift. The floor is a theorem about every positive actuation. The verdict stands: no actual dark energy crosses -1 . Criterion F1 names the measurement that would end it.

The value. The vacuum density is $(2.24 \text{ meV})^4$ (check B6). As a fraction of the Planck scale's fourth power it is $10^{-122.9}$, of the reduced Planck scale $10^{-120.1}$, and of the electroweak scale $10^{-56.2}$ (check B7). These are numbers. No seat reads a magnitude, so the value stands at the dot, measured and not derived.

9. WHAT THE DARK ROW CARRIES, AND WHAT IT DOES NOT

What it carries.

- The seat read on charge conjugation, with its fixed set carried to the line.
- Gravity as the one reading left, blind to identity.
- The record's forced half, the photon and Z nulls.
- The dark bit as the line property, spent by the act.
- The phantom refused by the floor.
- On the substrate, the dark matter on the line under both carriers.

What it does not carry.

- **The particle's identity:** gravity cannot read it, by theorem, and the identity row owes one non-gravitational detection.
- **The vacuum's value:** it stands at the dot.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

10. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F1, the phantom:** an equation of state below -1 at five sigma at any redshift, in two reconstructions that do not assume the two-parameter form, each on an independent supernova compilation, with the final DESI distances. It breaks the `the_phantom_is_negative_actuation` read physically, through the floor's premise that the dark energy's kinetic content is an actuation.
- **F2, a charge in the multiplet:** a charged partner of the dark matter, a tree-level photon, Z or gluon coupling, or a millicharge by any route. It breaks the placement on the fixed set (`dark_iff_fixed`, `visible_nulls_force_the_visible_fixed_set`).

- **F3, the dark bit set:** a dark self-interaction, dark radiation, a dark disk, or a dark matter distinct from its antiparticle, at five sigma, attributable to a dark charge. It breaks the spend of the `_dark_bit_is_the_line_property` and the `_dark_bit_is_the_majorana_bit`.

Owed. One non-gravitational detection of the dark matter, on the identity row. Nothing is owed on the seat's channel.

Ledger status. Row 2 gains the seat read on charge conjugation, WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains the dark-charged pair, the two species of one density, and the two closing worlds.

11. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens:

- $[\perp]$, a field sealed;
- $[\perp T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[.]$, the dot, which is no verdict.

Grades join at the weakest link. The engine's statements are theorem grade about the chart. The mapping of physical states onto them is structural grade. Every measured figure is corroboration grade. The dark bit's spend is at the root's grade, the act. $\Delta M = 0$.

The edition. Edition 1.0.0 (10.5281/zenodo.23006753, 27 September 2026) carried eighty-one theorems on a private staged chart and a thirty-five-check twin. This edition carries all eighty-one and adds forty-four:

- the hardware seat, category and functor (Part I);
- the energy carrier (Part II);
- the substrate (Part III);
- the dark row on the seat (Part XII);
- the capstone (Part XIII).

The private chart is retired. The thirty-five checks run unchanged beside three new ones on the substrate. The first edition's posit is now read as the line property and spent by the act. Its positioning table of prior candidates is replaced by the verdicts the theorems issue.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain D0 83f872725ad9 → D1 90fc702bbd1 → D2 492e519daab9 → D3 6618bec5164b (Appendix C).

The engine and the twin.

- **Engine.** SPHYS_Dark.lean, 1356 lines, SHA-256 466a1ef21f0ccca7..., one hundred twenty-five laws, no import and no axiom declared. Of the one hundred twenty-five, 42 depend on no axiom, 7 on propositional extensionality alone, 76 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen, and each law was negated in place and recompiled; the judgment is printed in Appendix C.
- **Twin.** Dark_Twin.f90, 381 lines, SHA-256 6e6d834fd9c6194e..., thirty-eight checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`. Its measured inputs enter as data with their sources named in its header.

Why an integer engine binds continuous physics. Charges are integers in sixths, by quantization, so the charge identities are exact. The anomaly conditions are sums over integer charges. The equation-of-state bounds are stated for every integer kinetic and potential term, and they scale to every rational. The twin checks the continuous quantities directly.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The dark matter is read at the Rest as the population that stands on the line and is seen by gravity alone. In the Lean codex's tokens its row renders $[\perp \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the companion papers of the hardware series, Immanent Gravity in particular;
- the first edition of this paper (10.5281/zenodo.23006753);
- PhysOS^T 1.0.5p in the public register (github.com/1000sapient/Trisduction, protocols/PhysOS/).

References.

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APPENDIX A: SPHYS_DARK.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Dark.lean · dark matter as the fixed set of charge conjugation,
the dark force as one keyed
bit, and the phantom as a negative actuation, read at the Riemann
seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared. Edition 2.0.0 of the dark
kernel: the content of Dark_Closure.lean (edition 1.0.0) re-seated on
the hardware seat, carriers
and substrate, its private stage chart retired.

The chart: h = 2 Re s, t = Im s; the fold s ↦ 1 - s is (h, t) ↦ (2 -
h, t), the critical line h = 1.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier (resident).
Part III  the substrate (resident).
Part IV   the fixed set and its readings.
Part V    the seat is not electric neutrality; a charge-free state
cannot radiate.
Part VI   gravity does not read identity.
Part VII  the neutrino fog: one energy record, two sources.
Part VIII the fifth reading, keyed by the anomaly conditions.
Part IX   the dark force in the acceleration sense: active density
and the kinetic floor.
Part X    the dark ledger, computed.
Part XI   fortification: the measured nulls force the visible fixed
set, the portals are
derived, the fifth is free, the dark bit is the Majorana
bit.
Part XII  the dark row on the seat.
Part XIII the capstone.
-/
set_option autoImplicit false
namespace SPHYS.Dark

/-- ## Part I. The seat, the category of involutions, and the functor of
the seat -/
abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_land (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/

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```

def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
    (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y}
  {p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
  Equivariant (g ∘ f) τ p := by
  intro x
  show g (f (τ x)) = p (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
by
  rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := (fold_involutive,
seat_fixed_line)

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involvement onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  (fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
    (fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=
fun _ hs => match hs with | (w, hw) => hw ▸ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▸ C.lands w

/-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/-! ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-! Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)

```

```

def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
    fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, - -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-! EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-! The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-! THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-! Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-! The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
phi (a, 0) ≠ p :=
  fun e => hp (e ▸ measured_on_line a)

/-! The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.ι w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-! ## Part III. The substrate: the particle map, three odd readings and
one even -/

/-! A particle in the broken phase: electric charge times three, colour,
baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-! The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-! The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := {c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror}
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

```

```

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((- q, - k, - b, - l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
by
  obtain (q, k, b, l, E) := c
  show ((- q, - k, - b, - l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat
c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun
c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
(leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-- ## Part IV. The fixed set and its readings. -/

/-- A state: hypercharge and weak isospin times six, one colour weight
(a Cartan coordinate), one dark charge, mass. -/
structure State where
  y6 : Int
  t3x6 : Int
  colour : Int
  dark : Int
  mass : Nat
  deriving DecidableEq, Repr

/-- Charge conjugation: every gauge charge flipped, the mass kept. -/
def conj (c : State) : State := {-c.y6, -c.t3x6, -c.colour, -c.dark,
c.mass}

/-- Visible conjugation: the three Standard Model charges flipped, the
dark charge and the mass kept. -/
def conjSM (c : State) : State := {-c.y6, -c.t3x6, -c.colour, c.dark,
c.mass}

def neutral (c : State) : Prop := c.y6 = 0 ∧ c.t3x6 = 0 ∧ c.colour = 0 ∧
c.dark = 0
def neutralSM (c : State) : Prop := c.y6 = 0 ∧ c.t3x6 = 0 ∧ c.colour = 0

theorem conj_involution (c : State) : conj (conj c) = c := by
  cases c with
  | mk y t k d m =>
    show State.mk (- y) (- t) (- k) (- d) m = State.mk y t k d m
    rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

/-- THE FIXED SET: a state is its own conjugate exactly when it carries
no gauge charge at all. -/
theorem fixed_iff_neutral (c : State) : conj c = c ↔ neutral c := by
  cases c with
  | mk y t k d m =>
    show State.mk (-y) (-t) (-k) (-d) m = State.mk y t k d m ↔ (y = 0 ∧
t = 0 ∧ k = 0 ∧ d = 0)
    constructor
    · intro h
      have h1 := congrArg State.y6 h
      have h2 := congrArg State.t3x6 h
      have h3 := congrArg State.colour h
      have h4 := congrArg State.dark h
      change -y = y at h1
      change -t = t at h2
      change -k = k at h3
      change -d = d at h4
      exact (by omega, by omega, by omega, by omega)
    · intro (h1, h2, h3, h4)
      subst h1; subst h2; subst h3; subst h4
      rfl

theorem fixedSM_iff_neutralSM (c : State) : conjSM c = c ↔ neutralSM c
:= by
  cases c with
  | mk y t k d m =>
    show State.mk (-y) (-t) (-k) d m = State.mk y t k d m ↔ (y = 0 ∧ t =
0 ∧ k = 0)
    constructor
    · intro h
      have h1 := congrArg State.y6 h
      have h2 := congrArg State.t3x6 h
      have h3 := congrArg State.colour h
      change -y = y at h1
      change -t = t at h2
      change -k = k at h3
      exact (by omega, by omega, by omega)
    · intro (h1, h2, h3)
      subst h1; subst h2; subst h3
      rfl

/-- The full fixed set lies inside the visible one. -/
theorem fixed_sub_fixedSM (c : State) (h : conj c = c) : conjSM c = c :=
by
  obtain (h1, h2, h3, _) := (fixed_iff_neutral c).mp h
  exact (fixedSM_iff_neutralSM c).mpr (h1, h2, h3)

/-- A dark-charged state: invisible to the three visible readings, read
by a fifth. -/
def darkCharged (m : Nat) : State := {0, 0, 0, 1, m}

theorem dark_charge_hidden_from_the_visible (m : Nat) :
  conjSM (darkCharged m) = darkCharged m ∧ conj (darkCharged m) ≠
darkCharged m := by
  refine (rfl, ?_)
  intro h
  have h4 : -(1 : Int) = 1 := congrArg State.dark h

```

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omega

/-- A reading is odd when conjugation flips its sign, even when
conjugation keeps it. -/
def OddReading (r : State → Int) : Prop := ∀ c, r (conj c) = -r c
def EvenReading {β : Type} (r : State → β) : Prop := ∀ c, r (conj c) =
r c

/-- EVERY ODD READING IS BLIND TO THE FIXED SET. -/
theorem odd_reading_blind (r : State → Int) (hr : OddReading r) (c :
State) (hc : conj c = c) :
  r c = 0 := by
  have h := hr c
  rw [hc] at h
  omega

/-- Every integer combination of the four charges is an odd reading. The
couplings of the photon, the Z, the
diagonal gluons and a dark photon to a state are all of this form.
-/
theorem charge_linear_readings_odd (a b k d : Int) :
  OddReading (fun c => a * c.y6 + b * c.t3x6 + k * c.colour + d *
c.dark) := by
  intro c
  show a * (-c.y6) + b * (-c.t3x6) + k * (-c.colour) + d * (-c.dark) =
-(a * c.y6 + b * c.t3x6 + k * c.colour + d * c.dark)
  simp only [Int.mul_neg]
  omega

/-- Electric charge times six, Q = T3 + Y. -/
def charge6 (c : State) : Int := c.t3x6 + c.y6

theorem charge_reading_odd : OddReading charge6 := by
  intro c
  show -c.t3x6 + -c.y6 = -(c.t3x6 + c.y6)
  omega

/-- Gravity reads the mass. -/
def readG (c : State) : Nat := c.mass

theorem gravity_reading_even : EvenReading readG := by
  intro c
  rfl

def darkState (m : Nat) : State := (0, 0, 0, 0, m)

theorem dark_state_fixed (m : Nat) : conj (darkState m) = darkState m :=
rfl

/-- ONLY GRAVITY READS THE FIXED SET: every odd reading returns zero on
it; the even reading returns the mass. -/
theorem only_gravity_reads_the_fixed_set (r : State → Int) (hr :
OddReading r) (m : Nat) :
  r (darkState m) = 0 ∧ readG (darkState m) = m :=
(oddd_reading_blind r hr (darkState m) (dark_state_fixed m), rfl)

/-- DARK IS FIXED: a state on which every odd reading returns zero is
exactly a state that conjugation fixes.
"Dark" in the gauge sense and "on the fixed set" are one property.
-/
theorem dark_iff_fixed (c : State) : (∀ r : State → Int, OddReading r
→ r c = 0) ↔ conj c = c := by
  constructor
  · intro h
    have h1 := h (fun x => x.y6) (fun _ => rfl)
    have h2 := h (fun x => x.t3x6) (fun _ => rfl)
    have h3 := h (fun x => x.colour) (fun _ => rfl)
    have h4 := h (fun x => x.dark) (fun _ => rfl)
    exact (fixed_iff_neutral c).mpr (h1, h2, h3, h4)
  · intro hc r hr
    exact odd_reading_blind r hr c hc

/-! ## Part V. The seat is not electric neutrality; a charge-free state
cannot radiate. -/

/-- A neutrino-like state: T3 = +1/2 and Y = -1/2, so electric charge
zero. The scalar partner of the neutrino
carries exactly these charges. -/
def sneutrinoLike (m : Nat) : State := (-3, 3, 0, 0, m)

theorem electric_neutrality_is_not_the_seat (m : Nat) :
  charge6 (sneutrinoLike m) = 0 ∧ conj (sneutrinoLike m) ≠
sneutrinoLike m := by
  refine (?, ?)
  · show (3 : Int) + -3 = 0
    decide
  · intro h
    have h1 := -(-3 : Int) = -3 := congrArg State.y6 h
    omega

theorem sq_nonneg (x : Int) : 0 ≤ x * x := by
  cases Int.le_total 0 x with
  | inl h => exact Int.mul_nonneg h h
  | inr h =>
    have hn : 0 ≤ -x := by omega
    have := Int.mul_nonneg hn hn
    rw [Int.neg_mul_neg] at this
    exact this

```

```

theorem sq_eq_zero {x : Int} (h : x * x = 0) : x = 0 := by
  cases Int.mul_eq_zero.mp h with
  | inl h => exact h
  | inr h => exact h

/-- The radiative weight: the sum of the squared charges, the strength
with which a state emits gauge quanta. -/
def radiative (c : State) : Int :=
  c.y6 * c.y6 + c.t3x6 * c.t3x6 + c.colour * c.colour + c.dark * c.dark

def radiativeSM (c : State) : Int := c.y6 * c.y6 + c.t3x6 * c.t3x6 +
c.colour * c.colour

theorem radiative_zero_iff_neutral (c : State) : radiative c = 0 ↔
neutral c := by
  cases c with
  | mk y t k d m =>
    show y * y + t * t + k * k + d * d = 0 ↔ (y = 0 ∧ t = 0 ∧ k = 0 ∧ d
= 0)
    have h1 := sq_nonneg y
    have h2 := sq_nonneg t
    have h3 := sq_nonneg k
    have h4 := sq_nonneg d
    constructor
    · intro h
      exact (sq_eq_zero (by omega), sq_eq_zero (by omega), sq_eq_zero
(by omega), sq_eq_zero (by omega))
    · intro (e1, e2, e3, e4)
      subst e1; subst e2; subst e3; subst e4
      decide

/-- NO CHARGE, NO RADIATION: the radiative weight vanishes exactly on
the fixed set. A state that cannot emit
cannot cool, and a population that cannot cool cannot settle into a
thin disk. -/
theorem no_charge_no_radiation (c : State) : radiative c = 0 ↔ conj c =
c :=
  (radiative_zero_iff_neutral c).trans (fixed_iff_neutral c).symm

/-- A dark-charged state emits no visible quanta and does emit dark
ones. -/
theorem dark_charged_radiates_only_dark (m : Nat) :
  radiativeSM (darkCharged m) = 0 ∧ radiative (darkCharged m) = 1 :=
by
  refine (?, ?)
  · show (0 : Int) * 0 + 0 * 0 + 0 * 0 = 0
    decide
  · show (0 : Int) * 0 + 0 * 0 + 0 * 0 + 1 * 1 = 1
    decide

/-- A multiplet is dark when every member sits on the fixed set. -/
def darkMultiplet (l : List State) : Bool := l.all (fun c => decide
(conj c = c))

/-- An isospin triplet with Y = 0: T3 = +1, 0, -1, times six. Its middle
member is neutral. -/
def tripletLike : List State := [(0, 6, 0, 0, 1), (0, 0, 0, 0, 1), (0,
-6, 0, 0, 1)]
def singletLike : List State := [(0, 0, 0, 0, 1)]

theorem a_neutral_member_is_not_a_dark_multiplet :
  tripletLike.any (fun c => decide (conj c = c)) = true ∧
darkMultiplet tripletLike = false ∧
darkMultiplet singletLike = true := by decide

/-! ## Part VI. Gravity does not read identity. -/

/-- A species: a label only a non-gravitational instrument reads, and
the mass density gravity reads. -/
structure Species where
  label : Nat
  density : Nat
  deriving DecidableEq, Repr

def gravRecord (s : Species) : Nat := s.density

/-- NO GRAVITATIONAL READING RETURNS THE IDENTITY: two species with one
density are one gravitational record. -/
theorem identity_unread (s t : Species) (hd : s.density = t.density) (hl
: s.label ≠ t.label)
  (g : Nat → Nat) : ¬ (g (gravRecord s) = s.label ∧ g (gravRecord t)
= t.label) := by
  intro (h1, h2)
  unfold gravRecord at h1 h2
  rw [hd] at h1
  exact hl (h1.symm.trans h2)

/-- The instance: a heavy particle and a light wave-like field, one
halo, one density. -/
def heavyHalo : Species := (1, 3)
def lightHalo : Species := (2, 3)

theorem two_species_one_record (g : Nat → Nat) :
  ¬ (g (gravRecord heavyHalo) = heavyHalo.label ∧ g (gravRecord
lightHalo) = lightHalo.label) :=
  identity_unread heavyHalo lightHalo rfl (by decide) g

```

```

/-! ## Part VII. The neutrino fog: one energy record, two sources. -/

/-- A nuclear recoil: its energy, and whether it points back to the Sun
or to the halo wind. -/
structure Recoil where
  energy : Nat
  solar : Bool
  deriving DecidableEq, Repr

/-- The direction flip exchanges the two sources and keeps the energy.
-/
def flipDir (r : Recoil) : Recoil := {r.energy, !r.solar}
def energyRecord (r : Recoil) : Nat := r.energy

theorem energy_record_even (r : Recoil) : energyRecord (flipDir r) =
  energyRecord r := rfl
theorem source_odd (r : Recoil) : (flipDir r).solar != r.solar := rfl

/-- THE FOG: at one energy, a solar recoil and a halo recoil leave one
energy record, and no reading of that
record returns the source. -/
theorem fog_one_record (g : Nat → Bool) (e : Nat) :
  ¬ (g (energyRecord (e, true)) = true ∧ g (energyRecord (e, false)) =
    false) := by
  intro (h1, h2)
  have e1 : g e = true := h1
  have e2 : g e = false := h2
  exact absurd (e1.symm.trans e2) (by decide)

/-- The direction reading separates them. -/
theorem direction_separates (e : Nat) : ((e, true) : Recoil).solar ≠
  ((e, false) : Recoil).solar := by
  intro h
  have h' : true = false := h
  exact absurd h' (by decide)

/-! ## Part VIII. The fifth reading. -/

/-- A left-handed Weyl field: multiplicity, hypercharge times six,
colour and isospin flags, a dark charge, and the
colour representation sign c3: +1 for a triplet, -1 for an
antitriplet, 0 for a colour singlet. -/
structure Weyl where
  name : String
  mult : Int
  y6 : Int
  triplet : Bool
  doublet : Bool
  qD : Int
  c3 : Int
  deriving Repr

/-- One Standard Model generation, no dark charge. -/
def sm : List Weyl :=
  [{"0", 6, 1, true, 0, 1}, {"u^c", 3, -4, true, false, 0, -1},
  {"d^c", 3, 2, true, false, 0, -1},
  {"L", 2, -3, false, true, 0, 0}, {"e^c", 1, 6, false, false, 0, 0}]

def total (f : Weyl → Int) (xs : List Weyl) : Int := xs.foldl (fun a w
=> a + f w) 0

/-- Anomaly freedom of one generation with a dark U(1): six visible
conditions and six dark ones (cubic,
gravitational, mixed with hypercharge twice, with colour and with
isospin). Each check is the anomaly coefficient up
to a positive normalization, Dynkin index 1/2 for every fundamental;
the colour cube is the representation sum with
A(3) = +1 and A(3bar) = -1. No check reads a field's name. -/
def closes (w : List Weyl) : Bool :=
  total (fun x => x.mult * x.y6) w == 0 &&
  total (fun x => x.mult * x.y6 ^ 3) w == 0 &&
  total (fun x => (x.mult / 3) * x.y6 (w.filter (·.triplet))) == 0 &&
  total (fun x => (x.mult / 2) * x.y6 (w.filter (·.doublet))) == 0 &&
  total (fun x => x.mult / 2 (w.filter (·.doublet))) % 2 == 0 &&
  total (fun x => (x.mult / 3) * x.c3 (w.filter (·.triplet))) == 0 &&
  total (fun x => x.mult * x.qD) w == 0 &&
  total (fun x => x.mult * (x.qD * x.qD * x.qD)) w == 0 &&
  total (fun x => x.mult * (x.qD * x.qD) * x.y6) w == 0 &&
  total (fun x => x.mult * x.qD * (x.y6 * x.y6)) w == 0 &&
  total (fun x => (x.mult / 3) * x.qD (w.filter (·.triplet))) == 0 &&
  total (fun x => (x.mult / 2) * x.qD (w.filter (·.doublet))) == 0

/-- A dark Weyl field: a Standard Model singlet with dark charge q. -/
def chi (q : Int) : Weyl := {"chi", 1, 0, false, false, q, 0}

/-- A vector-like dark pair without dark charge, and the same pair
charged +1 and -1. -/
def worldD0 : List Weyl := sm ++ [chi 0, chi 0]
def worldD : List Weyl := sm ++ [chi 1, chi (-1)]
/-- One chiral dark fermion of charge +1: anomalous. -/
def worldBad : List Weyl := sm ++ [chi 1]

/-- The fifth reading: some field carries a dark charge. -/
def fifth (w : List Weyl) : Bool := w.any (fun x => x.qD != 0)

/-- The template record: everything the visible closure reads, the dark
charge struck out. -/

```

```

def record (w : List Weyl) : List (String × Int × Int × Bool × Bool ×
Int) :=
  w.map (fun x => (x.name, x.mult, x.y6, x.triplet, x.doublet, x.c3))

theorem visible_generation_closes : closes sm = true := by decide
theorem closure_admits_both : closes worldD0 = true ∧ closes worldD =
true := by decide
theorem closure_is_not_vacuous : closes worldBad = false := by decide
theorem one_record_two_worlds : record worldD0 = record worldD := by
decide
theorem fifth_differs : fifth worldD0 = false ∧ fifth worldD = true :=
by decide

/-- NO TEMPLATE READING DECIDES THE FIFTH: no function of the template
record returns whether a dark charge
exists. -/
theorem no_template_reading_decides_fifth :
  ¬ ∃ g : List (String × Int × Int × Bool × Bool × Int) → Bool, ∀ w,
  g (record w) = fifth w := by
  intro (g, hg)
  have h1 := hg worldD0
  have h2 := hg worldD
  rw [one_record_two_worlds] at h1
  exact absurd (h1.symm.trans h2) (by decide)

/-- THE FIFTH IS KEYED: an anomaly-free world carries it and an
anomaly-free world lacks it. -/
theorem fifth_is_keyed :
  (∃ w, closes w = true ∧ fifth w = true) ∧ (∃ w, closes w = true ∧
  fifth w = false) :=
  ((worldD, by decide), (worldD0, by decide))

/-! ## Part IX. The dark force in the acceleration sense. -/

/-- The active gravitational density of a perfect fluid, rho + 3p. -/
def active (p p : Int) : Int := p + 3 * p

theorem dust_attracts (p : Int) (h : 0 < p) : 0 < active p 0 := by
  unfold active
  omega

theorem radiation_attracts (k : Int) (h : 0 < k) : 0 < active (3 * k) k
:= by
  unfold active
  omega

/-- VACUUM REPELS: pressure equal to minus the density gives active
density -2 rho. -/
theorem vacuum_repels (p : Int) (h : 0 < p) : active p (-p) = -2 * p ∧
active p (-p) < 0 := by
  unfold active
  constructor <=> omega

theorem repulsion_needs_tension (p p : Int) (h : active p p < 0) : 3 * p
< -p := by
  unfold active at h
  omega

/-- THE KINETIC FLOOR: a canonical field has rho = K + V and p = K - V
with K ≥ 0, so rho + p = 2K ≥ 0 and the
phantom divide w = -1 is never crossed from above. -/
theorem floor_forbids_phantom (K V : Int) (hK : 0 ≤ K) : 0 ≤ (K + V) +
(K - V) := by
  omega

/-- w = -1 exactly when the kinetic term vanishes: the frozen field. -/
theorem vacuum_iff_frozen (K V : Int) : (K + V) + (K - V) = 0 ↔ K = 0 :=
  (fun _ => by omega, fun _ => by omega)

theorem phantom_needs_ghost (K V : Int) (h : (K + V) + (K - V) < 0) : K
< 0 := by
  omega

/-- Twice the deceleration parameter of a flat matter plus vacuum
budget, in units of 1e-4. -/
def twoQ0 (om ol : Int) : Int := om - 2 * ol

/-- THE SIGN IS READ, NOT DERIVED: two flat budgets, one decelerating
and one accelerating; the measured
Planck 2018 budget is the second. -/
theorem flat_budgets_admit_both_signs :
  (10000 + 0 = (10000 : Int) ∧ 0 < twoQ0 10000 0) ∧
  (3153 + 6847 = (10000 : Int) ∧ twoQ0 3153 6847 < 0) := by decide

/-! ## Part X. The dark ledger, computed. -/

inductive Status
| crossed
| pending
| dot
deriving DecidableEq, Repr

def status (seatAbsent formalMass worldCarrier : Bool) : Status :=
  if seatAbsent then .dot else if formalMass || worldCarrier then
    .crossed else .pending

inductive Owed

```

```

| nothing
| oneCarrier
| oneOffering
deriving DecidableEq, Repr

def owedOf (st : Status) (tailEmpty : Bool) : Owed :=
  match st with
  | .pending => if tailEmpty then .oneOffering else .oneCarrier
  | _ => .nothing

structure Entry where
  name : String
  seatAbsent : Bool
  formalMass : Bool
  worldCarrier : Bool
  tailEmpty : Bool
  printed : Status
  owed : Owed

def darkLedger : List Entry :=
  [{"DM exists, read by gravity", false, false, true, false, .crossed,
    .nothing},
   {"DM identity and charges", false, false, false, false, .pending,
    .oneCarrier},
   {"a fifth odd reading, the dark gauge force", false, false, false,
    false, .pending, .oneCarrier},
   {"DE: sign of the acceleration", false, false, true, false, .crossed,
    .nothing},
   {"DE: w = -1 exactly, the frozen field", false, false, false, false,
    .pending, .oneCarrier},
   {"DE: magnitude of rho_Lambda, derived", false, false, false, true,
    .pending, .oneOffering}]

/-- Every printed status and every printed debt is the computed one. -/
theorem dark_ledger_is_computed :
  darkLedger.all (fun e => status e.seatAbsent e.formalMass
    e.worldCarrier == e.printed &&
    owedOf e.printed e.tailEmpty == e.owed) = true := by decide

theorem dark_ledger_counts :
  darkLedger.length = 6 ∧
  (darkLedger.filter (fun e => e.printed == .crossed)).length = 2 ∧
  (darkLedger.filter (fun e => e.printed == .pending)).length = 4 ∧
  (darkLedger.filter (fun e => e.printed == .dot)).length = 0 ∧
  (darkLedger.filter (fun e => e.owed == .oneCarrier)).length = 3 ∧
  (darkLedger.filter (fun e => e.owed == .oneOffering)).length = 1 :=
  by decide

/-- ## Audit additions, cycle darkforge, round 1 -/

/-- A C-fixed state is its mass: all four charges vanish. -/
theorem fixed_state_is_its_mass (c : State) (h : conj c = c) : c =
  darkState c.mass := by
  obtain (y, t, k, d, m) := c
  have hn : y = 0 ∧ t = 0 ∧ k = 0 ∧ d = 0 := (fixed_iff_neutral _).mp h
  obtain (rfl, rfl, rfl, rfl) := hn
  rfl

/-- Every reading, of either parity and any value type, returns on a
C-fixed state a function of its mass alone. -/
theorem every_template_reading_of_the_fixed_set_reads_only_mass {β :
Type} (r : State → β) (c : State)
  (h : conj c = c) : r c = r (darkState c.mass) :=
  congrArg r (fixed_state_is_its_mass c h)

/-- The Z reading in sixths, scaled by b: b T₃ minus a Q, the mixing
weight being the rational a / b. -/
def zReading (a b : Int) (c : State) : Int := b * c.t3x6 - a * charge6 c

theorem z_reading_odd (a b : Int) : OddReading (zReading a b) := by
  intro c
  show b * -c.t3x6 - a * (-c.t3x6 + -c.y6) = -(b * c.t3x6 - a * (c.t3x6 +
  c.y6))
  rw [Int.mul_add, Int.mul_add, Int.mul_neg, Int.mul_neg, Int.mul_neg]
  omega

/-- At zero electric charge the Z reading is b T₃, nonzero for every
rational weight a / b. -/
theorem z_reading_survives_neutrality (a b : Int) (m : Nat) (hb : b ≠ 0) :
  charge6 (sneutrinoLike m) = 0 ∧ zReading a b (sneutrinoLike m) = 3 *
  b ∧ 3 * b ≠ 0 := by
  refine ((by decide : (3 : Int) + -3 = 0), ?_, ?_)
  · show b * 3 - a * (3 + -3) = 3 * b
  · omega
  · omega

theorem the_triplet_has_one_fixed_member :
  (tripletLike.filter (fun c => decide (conj c = c))).length = 1 := by
  decide

/-- At positive density the floor is the bound w ≥ -1, for every
rational w = a/b with b > 0 and p = w p. -/
theorem floor_bounds_w (K V a b : Int) (hK : 0 ≤ K) (hp : 0 < K + V) (hb :
0 < b)
  (hw : b * (K - V) = a * (K + V)) : -b ≤ a := by
  have h2 : 0 ≤ b * ((K + V) + (K - V)) := Int.mul_nonneg (Int.le_of_lt
  hb) (by omega)

have h3 : b * (K - V) + b * (K + V) = b * ((K + V) + (K - V)) := by
  rw [Int.mul_add, Int.add_comm (K - V) (K + V)]
have h1 : (-b) * (K + V) ≤ a * (K + V) := by rw [Int.neg_mul, ← hw];
omega
exact Int.le_of_mul_le_mul_right h1 hp

/-- At negative density the floor does not bound w: K = 1 and V = -3
give p = -2, p = 4, w = -2. -/
theorem floor_needs_positive_density :
  ∃ K V a b : Int, 0 ≤ K ∧ K + V < 0 ∧ 0 < b ∧ b * (K - V) = a * (K +
  V) ∧ a < -b :=
  (1, -3, -2, 1, by decide, by decide, by decide, by decide)

/-- p + p summed over any list of fields (K, V). -/
def rhoPlusP : List (Int × Int) → Int
| [] => 0
| (k, v) :: t => (k + v) + (k - v) + rhoPlusP t

/-- The floor is additive: any number of fields with K ≥ 0 keeps p + p ≥
0. -/
theorem floor_is_additive (l : List (Int × Int)) (h : ∀ x ∈ l, 0 ≤ x.1) :
0 ≤ rhoPlusP l := by
  induction l with
  | nil => exact Int.le_refl 0
  | cons x t ih =>
    obtain (k, v) := x
    have hx : 0 ≤ k := h (k, v) (List.Mem.head t)
    have ht : 0 ≤ rhoPlusP t := ih (fun y hy => h y (List.Mem.tail (k,
    v) hy))
    show 0 ≤ (k + v) + (k - v) + rhoPlusP t
    omega

/-- The step from p + p ≥ 0 to w ≥ -1, for any totals, at positive
density, for every rational w = a/b, b > 0. -/
theorem w_floor_from_rho_plus_p (p p a b : Int) (h : 0 ≤ p + p) (hp : 0
< p) (hb : 0 < b)
  (hw : b * p = a * p) : -b ≤ a := by
  have h2 : 0 ≤ b * (p + p) := Int.mul_nonneg (Int.le_of_lt hb) h
  have h3 : b * p + b * p = b * (p + p) := by rw [Int.mul_add,
  Int.add_comm p p]
  have h1 : (-b) * p ≤ a * p := by rw [Int.neg_mul, ← hw]; omega
  exact Int.le_of_mul_le_mul_right h1 hp

def rhoTot : List (Int × Int) → Int
| [] => 0
| (k, v) :: t => (k + v) + rhoTot t

def pTot : List (Int × Int) → Int
| [] => 0
| (k, v) :: t => (k - v) + pTot t

theorem rhoPlusP_split (l : List (Int × Int)) : rhoPlusP l = rhoTot l +
pTot l := by
  induction l with
  | nil => rfl
  | cons x t ih =>
    obtain (k, v) := x
    show (k + v) + (k - v) + rhoPlusP t = ((k + v) + rhoTot t) + ((k -
    v) + pTot t)
    omega

/-- Any number of fields with K ≥ 0 at positive total density keeps w ≥
-1, for every rational w = a/b, b > 0. -/
theorem fields_bound_w (l : List (Int × Int)) (a b : Int) (h : ∀ x ∈ l,
0 ≤ x.1)
  (hp : 0 < rhoTot l) (hb : 0 < b) (hw : b * pTot l = a * rhoTot l) :
  -b ≤ a := by
  have h0 : 0 ≤ rhoTot l + pTot l := by
    have := floor_is_additive l h
    rw [rhoPlusP_split] at this
    exact this
  exact w_floor_from_rho_plus_p (rhoTot l) (pTot l) a b h0 hp hb hw

/-- ## Part XI. Fortification: the measured nulls force the fixed set,
the portals are derived, the fifth is free. -/

/-- The electric reading Q = T₃ + Y, times six. -/
def q6of (c : State) : Int := c.t3x6 + c.y6

/-- THE PHOTON AND THE Z FORCE THE ELECTROWEAK CHARGES TO ZERO. At any
mixing weight sin²θ W = a/b with b ≠ 0, a state
with Q = 0 and Z reading b.T₃ - a.Q = 0 has T₃ = 0 and Y = 0. -/
theorem photon_and_z_force_electroweak_zero (c : State) (a b : Int) (hb :
b ≠ 0)
  (hq : q6of c = 0) (hz : b * c.t3x6 - a * q6of c = 0) : c.t3x6 = 0 ∧
  c.y6 = 0 := by
  rw [hq, Int.mul_zero, Int.sub_zero] at hz
  have ht : c.t3x6 = 0 := by
    rcases Int.mul_eq_zero.mp hz with h | h
    · exact absurd h hb
    · exact h
  refine (ht, ?_)
  unfold q6of at hq
  omega

/-- THE VISIBLE NULLS FORCE THE VISIBLE FIXED SET: a colourless state
the photon and the Z both miss is fixed by the
```



```

Standard Model's conjugation. Nothing here reads the dark charge. -/
theorem visible_nulls_force_the_visible_fixed_set (c : State) (a b :
Int) (hb : b ≠ 0) (hq : q6of c = 0)
(hz : b * c.t3x6 - a * q6of c = 0) (hk : c.colour = 0) : conjSM c =
c := by
obtain (ht, hy) := photon_and_z_force_electroweak_zero c a b hb hq hz
exact (fixedSM_iff_neutralSM c).mpr (hy, ht, hk)

/-- QUANTIZATION TURNS A BOUND INTO A ZERO: an integer charge bounded
strictly inside one quantum is zero. -/
theorem below_one_quantum_is_zero (n : Int) (h1 : -1 < n) (h2 : n < 1) :
n = 0 := by omega

/-- THE MEASURED BOUNDS FORCE THE VISIBLE FIXED SET: with charges
quantized in sixths, a colourless state whose
electric and Z readings are each bounded strictly inside one quantum
is fixed by the Standard Model's conjugation. -/
theorem sub_quantum_bounds_force_the_visible_fixed_set (c : State) (a b :
Int) (hb : 0 < b)
(hq1 : -1 < q6of c) (hq2 : q6of c < 1)
(hz1 : -b < b * c.t3x6 - a * q6of c) (hz2 : b * c.t3x6 - a * q6of c
< b)
(hk : c.colour = 0) : conjSM c = c := by
have hq : q6of c = 0 := below_one_quantum_is_zero _ hq1 hq2
rw [hq, Int.mul_zero, Int.sub_zero] at hz1 hz2
have ht : c.t3x6 = 0 := by
rcases Int.lt_trichotomy c.t3x6 0 with h | h | h
· have : b * c.t3x6 ≤ b * (-1) := Int.mul_le_mul_of_nonneg_left (by
omega) (Int.le_of_lt hb)
omega
· exact h
· have : b * 1 ≤ b * c.t3x6 := Int.mul_le_mul_of_nonneg_left (by
omega) (Int.le_of_lt hb)
omega
exact visible_nulls_force_the_visible_fixed_set c a b (by omega) hq
(by rw [hq, ht]; simp) hk

/-- A field of the portal census: the five Standard Model left-handed
Weyl fermions, the Higgs doublet, a real singlet
scalar S and a singlet Weyl fermion N. -/
inductive Fld | Q | uc | dc | L | ec | H | S | N
deriving DecidableEq, Repr

def Fld.y6 : Fld → Int
| .Q => 1 | .uc => -4 | .dc => 2 | .L => -3 | .ec => 6 | .H => 3 | .S
=> 0 | .N => 0
def Fld.tri : Fld → Int
| .Q => 1 | .uc => -1 | .dc => -1 | _ => 0
def Fld.doublet : Fld → Bool
| .Q => true | .L => true | .H => true | _ => false
def Fld.darkF : Fld → Bool
| .S => true | .N => true | _ => false

/-- A leg of a monomial: a field and whether it enters conjugated. -/
abbrev Leg := Fld × Bool
def legY (l : Leg) : Int := if l.2 then -l.1.y6 else l.1.y6
def legTri (l : Leg) : Int := if l.2 then -l.1.tri else l.1.tri
def sumI : List Int → Int
| [] => 0
| x :: t => x + sumI t

/-- Gauge invariance of a monomial: hypercharge sums to zero, colour
triality to zero mod 3, and the doublets pair. -/
def invariantM (m : List Leg) : Bool :=
sumI (m.map legY) == 0 && sumI (m.map legTri) % 3 == 0 && (m.filter
(fun l => l.1.doublet)).length % 2 == 0

/-- A portal: an invariant monomial with a dark leg and a Standard Model
leg. -/
def portalM (m : List Leg) : Bool :=
invariantM m && m.any (fun l => l.1.darkF) && m.any (fun l =>
!l.1.darkF)

def scal : List Leg := [(H, false), (H, true), (S, false)]
def fer : List Fld := [Q, uc, dc, L, ec, N]

/-- Every renormalizable Lorentz-scalar monomial without derivatives:
two to four scalars, or two Weyl fermions of
one chirality with at most one scalar. 6 + 10 + 15 + 168 = 199
monomials. -/
def census : List (List Leg) :=
(List.range 3).flatMap (fun i => (List.range 3).flatMap (fun j =>
if i ≤ j then [(scal.getD i (S, false), scal.getD j (S, false))]
else [])) ++
(List.range 3).flatMap (fun i => (List.range 3).flatMap (fun j =>
(List.range 3).flatMap (fun k =>
if i ≤ j ∧ j ≤ k then [(scal.getD i (S, false), scal.getD j (S,
false), scal.getD k (S, false))]
else []))) ++
(List.range 3).flatMap (fun i => (List.range 3).flatMap (fun j =>
(List.range 3).flatMap (fun k =>
(List.range 3).flatMap (fun l => if i ≤ j ∧ j ≤ k ∧ k ≤ l then
[(scal.getD i (S, false), scal.getD j (S, false), scal.getD k
(S, false), scal.getD l (S, false))]
else [])))) ++
(List.range 6).flatMap (fun i => (List.range 6).flatMap (fun j =>
[false, true]).flatMap (fun cj =>
if i ≤ j then ([[]], [(H, false)], [(H, true)], [(S, false)]) :
List (List Leg)).map
(fun o => [(fer.getD i .N, cj), (fer.getD j .N, cj)] ++ o) else
[])))

set_option maxRecDepth 200000 in
/-- THE PORTALS ARE DERIVED: of the 199 renormalizable monomials of the
Standard Model with a singlet scalar and a
singlet fermion, exactly four couple the two sectors: S H†H, S² H†H,
and L N H with its conjugate. -/
theorem portals_are_derived :
census.length = 199 ∧ (census.filter portalM).length = 4 ∧
[(Fld.H, false), (Fld.H, true), (Fld.S, false)] ∈ census.filter
portalM ∧
[(Fld.H, false), (Fld.H, true), (Fld.S, false), (Fld.S, false)] ∈
census.filter portalM ∧
[(Fld.L, false), (Fld.N, false), (Fld.H, false)] ∈ census.filter
portalM ∧
[(Fld.L, true), (Fld.N, true), (Fld.H, true)] ∈ census.filter
portalM := by decide

/-- THE FIFTH IS FREE: a vector-like pair of Standard Model singlets
closes all twelve conditions whatever its dark
charge q, so every integer is allowed and the visible record fixes
none. -/
theorem fifth_is_free (q : Int) : closes (sm ++ [chi q, chi (-q)]) =
true := by
simp [closes, total, sm, chi, Int.neg_mul, Int.mul_neg,
Int.add_right_neg]

/-- THE TITLE'S TWO HALVES: C fixes a state exactly when the Standard
Model's conjugation fixes it and its dark charge
is zero. The first half is where measurement places dark matter; the
second is the one keyed bit. -/
theorem fixed_iff_visible_fixed_and_no_dark_charge (c : State) : conj c
= c ↔ (conjSM c = c ∧ c.dark = 0) := by
rw [fixed_iff_neutral, fixedSM_iff_neutralSM]
unfold neutral neutralSM
constructor
· intro (h1, h2, h3, h4); exact ((h1, h2, h3), h4)
· intro ((h1, h2, h3), h4); exact (h1, h2, h3, h4)

/-- The electric charge of each component of a field, times six: both
isospin components of a doublet. -/
def Fld.q6s : Fld → List Int
| .Q => [4, -2] | .uc => [-4] | .dc => [2] | .L => [0, -6] | .ec =>
[6] | .H => [6, 0] | .S => [0] | .N => [0]

/-- EVERY COLOURED STANDARD MODEL FIELD IS CHARGED in every component.
-/
theorem coloured_fields_are_charged : ∀ f : Fld, f.darkF = false →
f.tri ≠ 0 → f.q6s.all (· != 0) = true := by
intro f; cases f <|> decide

/-- A list of legs none of which is coloured has triality sum zero. -/
theorem sumI_tri_zero_of_none (rest : List Leg) (h : rest.any (fun l =>
l.1.tri != 0) = false) :
sumI (rest.map legTri) = 0 := by
induction rest with
| nil => rfl
| cons x t ih =>
have hx : (x.1.tri != 0) = false := by
cases hx' : (x.1.tri != 0) with
| false => rfl
| true => simp [List.any_cons, hx'] at h
have ht : t.any (fun l => l.1.tri != 0) = false := by
cases ht' : t.any (fun l => l.1.tri != 0) with
| false => rfl
| true => simp [List.any_cons, ht'] at h
have hx0 : x.1.tri = 0 := by simp using hx
show legTri x + sumI (t.map legTri) = 0
rw [ih ht]; unfold legTri; split <|> simp [hx0]

/-- A COLOURED RELIC BRINGS CHARGED PARTNERS: a colour-singlet composite
of one new colour triplet with Standard Model
legs contains a coloured Standard Model leg, and so an electrically
charged constituent. -/
theorem a_coloured_relic_brings_charged_partners (rest : List Leg) (hsm :
∀ l ∈ rest, l.1.darkF = false)
(hsing : (1 + sumI (rest.map legTri)) % 3 = 0) :
∃ l ∈ rest, l.1.tri ≠ 0 ∧ l.1.q6s.all (· != 0) = true := by
cases hb : rest.any (fun l => l.1.tri != 0) with
| false =>
have h0 := sumI_tri_zero_of_none rest hb
rw [h0] at hsing
exact absurd hsing (by decide)
| true =>
obtain (l, hl, hne) := List.any_eq_true.mp hb
have ht : l.1.tri ≠ 0 := by simp using hne
exact (l, hl, ht, coloured_fields_are_charged l.1 (hsm l hl) ht)

/-- The number of legs of one field in a monomial. -/
def darkCount (f : Fld) (m : List Leg) : Nat := (m.filter (fun l => l.1
== f)).length
/-- Even under the dark parity that keeps the dark matter stable: an
even number of S legs and of N legs. -/

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def evenDark (m : List Leg) : Bool := darkCount .S m % 2 == 0 &&
darkCount .N m % 2 == 0

set_option maxRecDepth 200000 in
/-- A STABLE SINGLET KEEPS ONE DOOR: under the parity that keeps the
dark matter stable exactly one portal survives,
S² HfH, and none survives for the fermion. -/
theorem a_stable_singlet_keeps_one_door :
(census.filter (fun m => portalM m && evenDark m)).length = 1 ∧
[[Fld.H, false), (Fld.H, true), (Fld.S, false), (Fld.S, false)] ∈
census.filter (fun m => portalM m && evenDark m) ∧
(census.filter (fun m => portalM m && evenDark m && (darkCount .N m
!= 0))).length = 0 := by decide

/-- A nonzero integer squares to at least one. -/
theorem one_le_sq_of_ne (q : Int) (h : q ≠ 0) : 1 ≤ q * q := by
rcases Int.lt_trichotomy q 0 with hq | hq | hq
· have h1 : q * q = (-q) * (-q) := (Int.neg_mul_neg q q).symm
have h2 : (-q) * 1 ≤ (-q) * (-q) := Int.mul_le_mul_of_nonneg_left
(by omega) (by omega)
omega
· exact absurd hq h
· have h2 : q * 1 ≤ q * q := Int.mul_le_mul_of_nonneg_left (by omega)
(by omega)
omega

/-- Every integer squares to at least zero. -/
theorem zero_le_sq (q : Int) : 0 ≤ q * q := by
rcases Int.lt_trichotomy q 0 with hq | hq | hq
· have := one_le_sq_of_ne q (by omega) ; omega
· subst hq; decide
· have := one_le_sq_of_ne q (by omega) ; omega

/-- A BOUND FIXES A CHARGE THROUGH A MEASURED COUPLING: with the
coupling fixed at g > 0, a force g q² below one
quantum's force g is the zero charge. This is the photon's case. -/
theorem measured_coupling_bound_forces_zero (g q : Int) (hg : 0 < g) (h :
g * (q * q) < g) : q = 0 := by
rcases Int.lt_trichotomy q 0 with hq | hq | hq
· have h2 : g * 1 ≤ g * (q * q) := Int.mul_le_mul_of_nonneg_left
(one_le_sq_of_ne q (by omega)) (by omega)
omega
· exact hq
· have h2 : g * 1 ≤ g * (q * q) := Int.mul_le_mul_of_nonneg_left
(one_le_sq_of_ne q (by omega)) (by omega)
omega

/-- A BOUND NEVER FIXES A CHARGE THROUGH A FREE COUPLING: for every
positive bound and every charge, some positive
coupling a/b puts the force (a/b) q² below the bound. This is the
dark photon's case. -/
theorem free_coupling_bound_never_forces_zero (B q : Int) (hB : 0 < B) :
∃ a b : Int, 0 < a ∧ 0 < b ∧ a * (q * q) < B * b := by
have h0 := zero_le_sq q
refine (1, q * q + 1, by decide, by omega, ?_)
have h1 : 1 * (q * q + 1) ≤ B * (q * q + 1) :=
Int.mul_le_mul_of_nonneg_right (by omega) (by omega)
omega

/-- The measured record of a state at mixing weight a/b: the photon
null, the Z null, colour neutral, positive mass. -/
def Admissible (a b : Int) (c : State) : Prop :=
q6of c = 0 ∧ b * c.t3x6 - a * q6of c = 0 ∧ c.colour = 0 ∧ 0 < c.mass

/-- THE FIRST CLAUSE IS FORCED: every state the measured record admits
is fixed by the Standard Model's conjugation. -/
theorem first_clause_is_forced (a b : Int) (hb : b ≠ 0) : ∀ c,
Admissible a b c → conjSM c = c :=
fun c (hq, hz, hk, _) => visible_nulls_force_the_visible_fixed_set c a
b hb hq hz hk

/-- THE SECOND CLAUSE IS KEYED: the measured record admits a world
without a dark charge and a world with one. -/
theorem second_clause_is_keyed (a b : Int) (m : Nat) (hm : 0 < m) :
Admissible a b (0, 0, 0, 0, m) ∧ Admissible a b (0, 0, 0, 1, m) :=
by
refine ((?_, ?_, rfl, hm), (?_, ?_, rfl, hm)) <|> simp [q6of]

/-- NO READING OF THE RECORD DECIDES THE BIT: no function of the visible
charges and the mass returns whether the dark
charge vanishes. -/
theorem no_reading_of_the_record_decides_the_bit (g : Int → Int → Int → Int → Bool) :
- ∀ c : State, g c.y6 c.t3x6 c.colour c.mass = decide (c.dark = 0)
:= by
intro h
have h1 : g 0 0 0 1 = true := h (0, 0, 0, 0, 1)
have h2 : g 0 0 0 1 = false := h (0, 0, 0, 1, 1)
exact Bool.noConfusion (h1.symm.trans h2)

/-- SELF-CONJUGACY SPENDS THE BIT: a state that is its own conjugate is
fixed by the Standard Model's conjugation and
carries no dark charge. -/
theorem self_conjugacy_spends_the_bit (c : State) (h : conj c = c) :
conjSM c = c ∧ c.dark = 0 :=
(fixed_iff_visible_fixed_and_no_dark_charge c).mp h

/-- THE DARK CLOSURE, EXECUTED: the visible fixed set, the forced half,
the division, the keyed bit, the two grades,
self-conjugacy, the crossing. A row closure; the dark row is not a
second seat. -/
theorem dark_closure_executed (a b : Int) (hb : b ≠ 0) :
(∀ c : State, conjSM (conjSM c) = c) ∧ (∀ c : State, conjSM c = c ↔
neutralSM c) ∧
(∀ c, Admissible a b c → conjSM c = c) ∧
(∀ c : State, conj c = c ↔ (conjSM c = c ∧ c.dark = 0)) ∧
(Admissible a b (0, 0, 0, 0, 1) ∧ Admissible a b (0, 0, 0, 1, 1)) ∧
(∀ g : Int → Int → Int → Nat → Bool, ¬ ∀ c : State, g c.y6
c.t3x6 c.colour c.mass = decide (c.dark = 0)) ∧
(∀ g q : Int, 0 < g → g * (q * q) < g → q = 0) ∧
(∀ B q : Int, 0 < B → ∃ a' b' : Int, 0 < a' ∧ 0 < b' ∧ a' * (q * q)
< B * b') ∧
(∀ c : State, conj c = c → conjSM c = c ∧ c.dark = 0) ∧
(∀ (Q : Prop) (c : State), Q → ((Q → c.dark = 0) ↔ c.dark = 0)) :=
{ (fun c => by cases c; simp [conjSM], fixedSM_iff_neutralSM),
first_clause_is_forced a b hb,
fixed_iff_visible_fixed_and_no_dark_charge, second_clause_is_keyed a
b 1 (by decide),
no_reading_of_the_record_decides_the_bit,
measured_coupling_bound_forces_zero,
free_coupling_bound_never_forces_zero, self_conjugacy_spends_the_bit,
fun _ _ hq => (fun f => f hq, fun h _ => h))

/-- The abelian charges a self-paired mass term carries: a Majorana mass
ψψ, or the mass S² of a real scalar, pairs a
field with itself, so it carries twice its hypercharge and twice its
dark charge. -/
def selfPairCharges (c : State) : Int × Int := (2 * c.y6, 2 * c.dark)
/-- A self-paired mass term is allowed when it carries no abelian
charge. -/
def MajoranaAllowed (c : State) : Prop := selfPairCharges c = (0, 0)

/-- A MAJORANA MASS FORBIDS EVERY ABELIAN CHARGE: a self-paired mass
term is allowed exactly when the hypercharge and
the dark charge both vanish. The route is the mass term, not the
definition of conjugation. -/
theorem majorana_mass_forbids_abelian_charge (c : State) :
MajoranaAllowed c ↔ (c.y6 = 0 ∧ c.dark = 0) := by
unfold MajoranaAllowed selfPairCharges
constructor
· intro h
have h1 : 2 * c.y6 = 0 := congrArg Prod.fst h
have h2 : 2 * c.dark = 0 := congrArg Prod.snd h
exact (by omega, by omega)
· intro (h1, h2)
simp [h1, h2]

/-- THE DARK BIT IS THE MAJORANA BIT: on the visible fixed set, a
self-paired mass is allowed exactly when the dark
charge is zero, and exactly when conjugation fixes the state. -/
theorem the_dark_bit_is_the_majorana_bit (c : State) (h : conjSM c = c) :
(MajoranaAllowed c ↔ c.dark = 0) ∧ (MajoranaAllowed c ↔ conj c = c)
:= by
have hy : c.y6 = 0 := ((fixedSM_iff_neutralSM c).mp h).1
constructor
· rw [majorana_mass_forbids_abelian_charge]
exact (fun h' => h'.2, fun h' => {hy, h'})
· rw [majorana_mass_forbids_abelian_charge,
fixed_iff_visible_fixed_and_no_dark_charge]
exact (fun h' => {h, h'.2}, fun h' => {hy, h'.2})

/-- A DARK CHARGE MAKES A DIRAC PAIR: a mass term pairing two states is
neutral in the dark charge only if their dark
charges cancel, so a dark-charged state takes its mass with a
distinct partner. -/
theorem a_dark_charge_makes_a_dirac_pair (c c' : State) (hpair : c.dark +
c'.dark = 0) (hd : c.dark ≠ 0) :
c'.dark = -c.dark ∧ c' ≠ c := by
refine (by omega, fun he => hd ?_)
have : c'.dark = c.dark := by rw [he]
omega

/-- ## Part XII. The dark row on the seat -/

/-- The dark carrier: the dark charge is the offset from the line, the
mass the height. -/
def darkSeat (c : State) : Pt := (1 + c.dark, (c.mass : Int))

/-- THE DARK CARRIER IS EQUIVARIANT: charge conjugation is carried onto
the fold. -/
theorem dark_carrier_equivariant : Equivariant darkSeat conj fold := by
intro c
show ((1 + -c.dark, (c.mass : Int)) : Pt) = (2 - (1 + c.dark), (c.mass : Int))
rw [pe]; exact (by omega, rfl)

/-- The landing is forced: the fixed set of conjugation is carried onto
the line by the functor. -/
theorem dark_carrier_lands :
∃ C : Carrier { c : State // conj c = c }, ∀ w, C.l w = darkSeat w.1
:=

```

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equivariant_carrier_lands darkSeat conj dark_carrier_equivariant

/- THE DARK BIT IS THE LINE PROPERTY: on the visible fixed set a state
lands on the line exactly
when its dark charge is zero, exactly when conjugation fixes it. -/
theorem the_dark_bit_is_the_line_property (c : State) (h : conjSM c = c) :
:
  (OnLine (darkSeat c) ↔ c.dark = 0) ∧ (OnLine (darkSeat c) ↔ conj c =
  c) := by
  have e : OnLine (darkSeat c) ↔ c.dark = 0 := by
  show (1 + c.dark = 1) ↔ c.dark = 0
  exact (fun h1 => by omega, fun h1 => by omega)
  refine (e, e.trans ?_)
  rw [fixed_iff_visible_fixed_and_no_dark_charge]
  exact (fun hd => (h, hd), fun hh => hh.2)

/- A DARK-CHARGED PAIR IS AN OFF-LINE FOLD PAIR WITH ONE RECORD: a
dark-charged state and its
conjugate land on two distinct points off the line, exchanged by the
fold, at one height, and
the registration keeps the height and forgets the side. -/
theorem a_dark_pair_is_an_off_line_pair (c : State) (hd : c.dark ≠ 0) :
darkSeat (conj c) = fold (darkSeat c) ∧ darkSeat (conj c) ≠ darkSeat
c ∧
  ~ OnLine (darkSeat c) ∧ reg (darkSeat (conj c)) = reg (darkSeat c)
:= by
  refine (dark_carrier_equivariant c, fun he => hd ?_, fun hl => hd ?_,
  ?_)
  · have h1 : (darkSeat (conj c)).1 = (darkSeat c).1 := by rw [he]
  have h2 : 1 + ~c.dark = 1 + c.dark := h1
  omega
  · have h2 : 1 + c.dark = 1 := h1
  omega
  · rw [dark_carrier_equivariant c]; exact reg_forgets_side (darkSeat c)

/- DARK MATTER STANDS ON THE LINE TWICE: on the substrate, a neutral
stable particle is fixed by both
involutions and lands on the line under the charge carrier and under
the energy carrier. -/
theorem dark_matter_on_the_line_twice (c : Particle) (hN : Neutral c)
(hS : IsReal c.energy) :
  (bar c = c ∧ tmirror c = c) ∧ OnLine (chargeSeat c) ∧ OnLine
  (energySeat c) :=
  ((joint_fixed_iff c).mpr (hN, hS), neutral_charge_on_line c hN.1,
  (phi_line_iff_real c.energy).mpr hS)

/- GRAVITY READS THE FIXED SET AND NOTHING ELSE: every gravitational
reading of a particle and its
conjugate agrees, and on a conjugation-fixed state every template
reading returns a function of
the mass alone. -/
theorem gravity_is_the_reading_left {β : Type} (g : Energy → β) (c :
Particle) (r : State → β) (d : State)
(h : conj d = d) :
  g (bar c).energy = g c.energy ∧ r d = r (darkState d.mass) :=
  (gravity_reads_no_charge_bit g c,
  every_template_reading_of_the_fixed_set_reads_only_mass r d h)

/- THE PHANTOM IS A NEGATIVE ACTUATION: an equation of state below -1
needs a negative kinetic
term, which the actuation floor refuses; at the floor the vacuum is
frozen kinetic content. -/
theorem the_phantom_is_negative_actuation (K V : Int) :
  ((K + V) + (K - V) < 0 → K < 0) ∧ (0 ≤ K → 0 ≤ (K + V) + (K - V))
  ∧
  ((K + V) + (K - V) = 0 ↔ K = 0) :=
  (phantom_needs_ghost K V, floor_forbids_phantom K V, vacuum_iff_frozen
  K V)

/-! ## Part XIII. The capstone -/

/- DARK MATTER IS THE FIXED SET, THE DARK FORCE ONE KEYED BIT, AND THE
PHANTOM A NEGATIVE ACTUATION.
Charge conjugation is carried onto the fold and its fixed set onto
the line; every odd reading is
blind there, so gravity is the reading left; the record forces the
visible half and keys the dark
half, which is the line property carried; a dark-charged pair is an
off-line fold pair with one
record; a neutral stable particle stands on the line under both
carriers; and the actuation floor
forbids the phantom. -/
theorem dark_matter_is_the_fixed_set :
  Equivariant darkSeat conj fold ∧
  (V c : State, conj c = c ↔ neutral c) ∧
  (V (r : State → Int), OddReading r → ∀ c, conj c = c → r c = 0) ∧
  (V c : State, conj c = c ↔ (conjSM c = c ∧ c.dark = 0)) ∧
  (V c : State, conjSM c = c → (OnLine (darkSeat c) ↔ conj c = c)) ∧
  (V c : State, c.dark ≠ 0 → ~ OnLine (darkSeat c) ∧ reg (darkSeat
  (conj c)) = reg (darkSeat c)) ∧
  (V c : Particle, Neutral c → IsReal c.energy → OnLine (chargeSeat
  c) ∧ OnLine (energySeat c)) ∧
  (V K V : Int, (K + V) + (K - V) < 0 → K < 0) :=
  (dark_carrier_equivariant, fixed_iff_neutral, fun r hr c hc =>
  odd_reading_blind r hr c hc,
  fixed_iff_visible_fixed_and_no_dark_charge,
  fun c h => (the_dark_bit_is_the_line_property c h).2,

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fun c hd => ((a_dark_pair_is_an_off_line_pair c hd).2.2.1,
(a_dark_pair_is_an_off_line_pair c hd).2.2.2),
fun c hN hS => ((dark_matter_on_the_line_twice c hN hS).2.1,
(dark_matter_on_the_line_twice c hN hS).2.2),
phantom_needs_ghost)

end SPHYS.Dark

#print axioms SPHYS.Dark.pe
#print axioms SPHYS.Dark.fold_involutive
#print axioms SPHYS.Dark.seat_fixed_line
#print axioms SPHYS.Dark.reg_lands
#print axioms SPHYS.Dark.reg_fixes_iff
#print axioms SPHYS.Dark.reg_forgets_side
#print axioms SPHYS.Dark.least_erasure_iff_value
#print axioms SPHYS.Dark.equivariant_id
#print axioms SPHYS.Dark.equivariant_comp
#print axioms SPHYS.Dark.fix_functorial
#print axioms SPHYS.Dark.fold_global_seat
#print axioms SPHYS.Dark.equivariant_carrier_lands
#print axioms SPHYS.Dark.value_on_image
#print axioms SPHYS.Dark.kinetic_crossing
#print axioms SPHYS.Dark.off_locus_pair
#print axioms SPHYS.Dark.ee
#print axioms SPHYS.Dark.mirror_involutive
#print axioms SPHYS.Dark.psi_phi
#print axioms SPHYS.Dark.phi_psi
#print axioms SPHYS.Dark.phi_injective
#print axioms SPHYS.Dark.phi_equivariant
#print axioms SPHYS.Dark.mirror_fixed_iff_real
#print axioms SPHYS.Dark.phi_line_iff_real
#print axioms SPHYS.Dark.stationary_iff_real
#print axioms SPHYS.Dark.line_is_stationary
#print axioms SPHYS.Dark.measured_on_line
#print axioms SPHYS.Dark.no_measurement_off_line
#print axioms SPHYS.Dark.energy_carrier_lands
#print axioms SPHYS.Dark.bar_involutive
#print axioms SPHYS.Dark.bar_fixed_iff_neutral
#print axioms SPHYS.Dark.odd_charges_even_energy
#print axioms SPHYS.Dark.gravity_reads_no_charge_bit
#print axioms SPHYS.Dark.involutions_commute
#print axioms SPHYS.Dark.tmirror_involutive
#print axioms SPHYS.Dark.joint_fixed_iff
#print axioms SPHYS.Dark.charge_carrier_equivariant
#print axioms SPHYS.Dark.pair_lands_once
#print axioms SPHYS.Dark.tmirror_equivariant
#print axioms SPHYS.Dark.neutral_charge_on_line
#print axioms SPHYS.Dark.pair_on_the_edges
#print axioms SPHYS.Dark.stable_lands
#print axioms SPHYS.Dark.conj_involution
#print axioms SPHYS.Dark.fixed_iff_neutral
#print axioms SPHYS.Dark.fixedSM_iff_neutralSM
#print axioms SPHYS.Dark.fixed_sub_fixedSM
#print axioms SPHYS.Dark.dark_charge_hidden_from_the_visible
#print axioms SPHYS.Dark.odd_reading_blind
#print axioms SPHYS.Dark.charge_linear_readings_odd
#print axioms SPHYS.Dark.charge_reading_odd
#print axioms SPHYS.Dark.gravity_reading_even
#print axioms SPHYS.Dark.dark_state_fixed
#print axioms SPHYS.Dark.only_gravity_reads_the_fixed_set
#print axioms SPHYS.Dark.dark_iff_fixed
#print axioms SPHYS.Dark.electric_neutrality_is_not_the_seat
#print axioms SPHYS.Dark.sq_nonneg
#print axioms SPHYS.Dark.sq_eq_zero
#print axioms SPHYS.Dark.radiative_zero_iff_neutral
#print axioms SPHYS.Dark.no_charge_no_radiation
#print axioms SPHYS.Dark.dark_charged_radiates_only_dark
#print axioms SPHYS.Dark.a_neutral_member_is_not_a_dark_multiset
#print axioms SPHYS.Dark.identity_unread
#print axioms SPHYS.Dark.two_species_one_record
#print axioms SPHYS.Dark.energy_record_even
#print axioms SPHYS.Dark.source_odd
#print axioms SPHYS.Dark.fog_one_record
#print axioms SPHYS.Dark.direction_separates
#print axioms SPHYS.Dark.visible_generation_closes
#print axioms SPHYS.Dark.closure_admits_both
#print axioms SPHYS.Dark.closure_is_not_vacuous
#print axioms SPHYS.Dark.one_record_two_worlds
#print axioms SPHYS.Dark.fifth_differs
#print axioms SPHYS.Dark.no_template_reading_decides_fifth
#print axioms SPHYS.Dark.fifth_is_keyed
#print axioms SPHYS.Dark.dust_attracts
#print axioms SPHYS.Dark.radiation_attracts
#print axioms SPHYS.Dark.vacuum_repels
#print axioms SPHYS.Dark.repulsion_needs_tension
#print axioms SPHYS.Dark.floor_forbids_phantom
#print axioms SPHYS.Dark.vacuum_iff_frozen
#print axioms SPHYS.Dark.phantom_needs_ghost
#print axioms SPHYS.Dark.flat_budgets_admit_both_signs
#print axioms SPHYS.Dark.dark_ledger_is_computed
#print axioms SPHYS.Dark.dark_ledger_counts
#print axioms SPHYS.Dark.fixed_state_is_its_mass
#print axioms
SPHYS.Dark.every_template_reading_of_the_fixed_set_reads_only_mass
#print axioms SPHYS.Dark.z_reading_odd
#print axioms SPHYS.Dark.z_reading_survives_neutrality
#print axioms SPHYS.Dark.the_triplet_has_one_fixed_member
#print axioms SPHYS.Dark.floor_bounds_w

```

```
#print axioms SPHYS.Dark.floor_needs_positive_density
#print axioms SPHYS.Dark.floor_is_additive
#print axioms SPHYS.Dark.w_floor_from_rho_plus_p
#print axioms SPHYS.Dark.rhoPlusP_split
#print axioms SPHYS.Dark.fields_bound_w
#print axioms SPHYS.Dark.photon_and_z_force_electroweak_zero
#print axioms SPHYS.Dark.visible_nulls_force_the_visible_fixed_set
#print axioms SPHYS.Dark.below_one_quantum_is_zero
#print axioms SPHYS.Dark.sub_quantum_bounds_force_the_visible_fixed_set
#print axioms SPHYS.Dark.portals_are_derived
#print axioms SPHYS.Dark.fifth_is_free
#print axioms SPHYS.Dark.fixed_iff_visible_fixed_and_no_dark_charge
#print axioms SPHYS.Dark.coloured_fields_are_charged
#print axioms SPHYS.Dark.sumI_tri_zero_of_none
#print axioms SPHYS.Dark.a_coloured_relic_brings_charged_partners
#print axioms SPHYS.Dark.a_stable_singlet_keeps_one_door
#print axioms SPHYS.Dark.one_le_sq_of_ne
#print axioms SPHYS.Dark.zero_le_sq
```

```
'SPHYS.Dark.pe' does not depend on any axioms
'SPHYS.Dark.fold_involutive' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.seat_fixed_line' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.reg_lands' does not depend on any axioms
'SPHYS.Dark.reg_fixes_iff' depends on axioms: [proprext]
'SPHYS.Dark.reg_forgets_side' does not depend on any axioms
'SPHYS.Dark.least_eraser_iff_value' depends on axioms: [proprext]
'SPHYS.Dark.equivariant_id' does not depend on any axioms
'SPHYS.Dark.equivariant_comp' does not depend on any axioms
'SPHYS.Dark.fix_functorial' does not depend on any axioms
'SPHYS.Dark.fold_global_seat' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.equivariant_carrier_lands' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.value_on_image' does not depend on any axioms
'SPHYS.Dark.kinetic_crossing' does not depend on any axioms
'SPHYS.Dark.off_locus_pair' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.ee' does not depend on any axioms
'SPHYS.Dark.mirror_involutive' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.psi_phi' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.phi_psi' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.phi_injective' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.phi_equivariant' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.mirror_fixed_iff_real' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.phi_line_iff_real' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.stationary_iff_real' depends on axioms: [proprext]
'SPHYS.Dark.line_is_stationary' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.measured_on_line' does not depend on any axioms
'SPHYS.Dark.no_measurement_off_line' does not depend on any axioms
'SPHYS.Dark.energy_carrier_lands' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.bar_involutive' does not depend on any axioms
'SPHYS.Dark.bar_fixed_iff_neutral' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.odd_charges_even_energy' does not depend on any axioms
'SPHYS.Dark.gravity_reads_no_charge_bit' does not depend on any axioms
'SPHYS.Dark.involutions_commute' does not depend on any axioms
'SPHYS.Dark.tmirror_involutive' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.joint_fixed_iff' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.charge_carrier_equivariant' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.pair_lands_once' does not depend on any axioms
'SPHYS.Dark.tmirror_equivariant' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.neutral_charge_on_line' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.pair_on_the_edges' does not depend on any axioms
'SPHYS.Dark.stable_lands' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.conj_involution' does not depend on any axioms
'SPHYS.Dark.fixed_iff_neutral' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.fixedSM_iff_neutralSM' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.fixed_sub_fixedSM' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.dark_charge_hidden_from_the_visible' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.odd_reading_blind' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.charge_linear_readings_odd' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.charge_reading_odd' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.gravity_reading_even' does not depend on any axioms
'SPHYS.Dark.dark_state_fixed' does not depend on any axioms
'SPHYS.Dark.only_gravity_reads_the_fixed_set' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.dark_iff_fixed' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.electric_neutrality_is_not_the_seat' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.sq_nonneg' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.sq_eq_zero' depends on axioms: [proprext]
'SPHYS.Dark.radiative_zero_iff_neutral' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.no_charge_no_radiation' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.dark_charged_radiates_only_dark' does not depend on any axioms
'SPHYS.Dark.a_neutral_member_is_not_a_dark_multiplet' does not depend on any axioms
'SPHYS.Dark.identity_unread' does not depend on any axioms
```

```
#print axioms SPHYS.Dark.measured_coupling_bound_forces_zero
#print axioms SPHYS.Dark.free_coupling_bound_never_forces_zero
#print axioms SPHYS.Dark.first_clause_is_forced
#print axioms SPHYS.Dark.second_clause_is_keyed
#print axioms SPHYS.Dark.no_reading_of_the_record_decides_the_bit
#print axioms SPHYS.Dark.self_conjugacy_spends_the_bit
#print axioms SPHYS.Dark.dark_closure_executed
#print axioms SPHYS.Dark.majorana_mass_forbids_abelian_charge
#print axioms SPHYS.Dark.the_dark_bit_is_the_majorana_bit
#print axioms SPHYS.Dark.a_dark_charge_makes_a_dirac_pair
#print axioms SPHYS.Dark.dark_carrier_equivariant
#print axioms SPHYS.Dark.dark_carrier_lands
#print axioms SPHYS.Dark.the_dark_bit_is_the_line_property
#print axioms SPHYS.Dark.a_dark_pair_is_an_off_line_pair
#print axioms SPHYS.Dark.dark_matter_on_the_line_twice
#print axioms SPHYS.Dark.gravity_is_the_reading_left
#print axioms SPHYS.Dark.the_phantom_is_negative_actuation
#print axioms SPHYS.Dark.dark_matter_is_the_fixed_set
```

```
'SPHYS.Dark.two_species_one_record' does not depend on any axioms
'SPHYS.Dark.energy_record_even' does not depend on any axioms
'SPHYS.Dark.source_odd' does not depend on any axioms
'SPHYS.Dark.fog_one_record' does not depend on any axioms
'SPHYS.Dark.direction_separates' does not depend on any axioms
'SPHYS.Dark.visible_generation_closes' does not depend on any axioms
'SPHYS.Dark.closure_admits_both' does not depend on any axioms
'SPHYS.Dark.closure_is_not_vacuous' does not depend on any axioms
'SPHYS.Dark.one_record_two_worlds' does not depend on any axioms
'SPHYS.Dark.fifth_differs' does not depend on any axioms
'SPHYS.Dark.no_template_reading_decides_fifth' does not depend on any axioms
'SPHYS.Dark.fifth_is_keyed' does not depend on any axioms
'SPHYS.Dark.dust_attracts' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.radiation_attracts' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.vacuum_repels' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.repulsion_needs_tension' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.floor_forbids_phantom' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.vacuum_iff_frozen' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.phantom_needs_ghost' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.flat_budgets_admit_both_signs' does not depend on any axioms
'SPHYS.Dark.dark_ledger_is_computed' does not depend on any axioms
'SPHYS.Dark.dark_ledger_counts' does not depend on any axioms
'SPHYS.Dark.fixed_state_is_its_mass' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.every_template_reading_of_the_fixed_set_reads_only_mass' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.z_reading_odd' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.z_reading_survives_neutrality' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.the_triplet_has_one_fixed_member' does not depend on any axioms
'SPHYS.Dark.floor_bounds_w' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.floor_needs_positive_density' does not depend on any axioms
'SPHYS.Dark.floor_is_additive' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.w_floor_from_rho_plus_p' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.rhoPlusP_split' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.fields_bound_w' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.photon_and_z_force_electroweak_zero' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.visible_nulls_force_the_visible_fixed_set' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.below_one_quantum_is_zero' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.sub_quantum_bounds_force_the_visible_fixed_set' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.portals_are_derived' depends on axioms: [proprext]
'SPHYS.Dark.fifth_is_free' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.fixed_iff_visible_fixed_and_no_dark_charge' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.coloured_fields_are_charged' does not depend on any axioms
'SPHYS.Dark.sumI_tri_zero_of_none' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.a_coloured_relic_brings_charged_partners' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.a_stable_singlet_keeps_one_door' depends on axioms: [proprext]
'SPHYS.Dark.one_le_sq_of_ne' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.zero_le_sq' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.measured_coupling_bound_forces_zero' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.free_coupling_bound_never_forces_zero' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.first_clause_is_forced' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.second_clause_is_keyed' depends on axioms: [proprext]
'SPHYS.Dark.no_reading_of_the_record_decides_the_bit' does not depend on any axioms
'SPHYS.Dark.self_conjugacy_spends_the_bit' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.dark_closure_executed' depends on axioms: [proprext, Quot.sound]
'SPHYS.Dark.majorana_mass_forbids_abelian_charge' depends on axioms: [proprext, Quot.sound]
```

```
'SPHYS.Dark.the_dark_bit_is_the_majorana_bit' depends on axioms:
[propept, Quot.sound]
'SPHYS.Dark.a_dark_charge_makes_a_dirac_pair' depends on axioms:
[propept, Quot.sound]
'SPHYS.Dark.dark_carrier_equivariant' depends on axioms: [propept,
Quot.sound]
'SPHYS.Dark.dark_carrier_lands' depends on axioms: [propept, Quot.sound]
'SPHYS.Dark.the_dark_bit_is_the_line_property' depends on axioms:
[propept, Quot.sound]
```

```
'SPHYS.Dark.a_dark_pair_is_an_off_line_pair' depends on axioms:
[propept, Quot.sound]
'SPHYS.Dark.dark_matter_on_the_line_twice' depends on axioms: [propept,
Quot.sound]
'SPHYS.Dark.gravity_is_the_reading_left' depends on axioms: [propept,
Quot.sound]
'SPHYS.Dark.the_phantom_is_negative_actuation' depends on axioms:
[propept, Quot.sound]
'SPHYS.Dark.dark_matter_is_the_fixed_set' depends on axioms: [propept,
Quot.sound]
```

APPENDIX B: DARK_TWIN.F90 AND ITS RUN

```
! Dark_Twin.f90 · edition 2.0.0 · the executed twin of SPHYS_Dark.lean ·
Fortran 2018.
! Edition 1.0.0's thirty-five checks carried unchanged; block S added,
the dark row on the hardware seat.
! The dark sector on the closure template: the fixed set and its
readings on a grid, the radiative weight, the
! multiplet test, the anomaly sums of the dark worlds, the measured
budget, the kinetic floor, the phantom
! crossing of the fitted CPL curve, the geometry of the neutrino fog,
the dark-force floors, and one halo read
! by two species. Every figure is computed in this run; cited inputs
enter as data with their source named.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Dark_Twin.f90
program dark_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
real(dp), parameter :: PI = 3.14159265358979323846_dp, DEG =
PI/180.0_dp
! Planck 2018, TT,TE,EE+lowE+lensing (Planck Collaboration VI 2020,
A&A 641, A6, Table 2)
real(dp), parameter :: OMC_H2 = 0.1200_dp, OMB_H2 = 0.02237_dp, H100 =
0.6736_dp
real(dp), parameter :: OM_L = 0.6847_dp, OM_M = 0.3153_dp, SIG_OM =
0.0073_dp
real(dp), parameter :: SUM_MNU = 0.06_dp ! eV,
the minimal sum Planck assumes
! CODATA 2018 and IAU 2012
real(dp), parameter :: G = 6.67430e-11_dp, C = 299792458.0_dp, MPC =
3.0856775814913673e22_dp
real(dp), parameter :: EV = 1.602176634e-19_dp, HBARC =
1.973269804e-7_dp, KB = 1.380649e-23_dp
real(dp), parameter :: MPL = 1.220890e28_dp, MPLRED = 2.435e27_dp, VEW =
246.22e9_dp ! eV
real(dp), parameter :: GEV_G = 1.78266192e-24_dp
! grams per GeV/c^2
! DESI DR2 BAO + CMB + DESY5, CPL best fit (DESI Collaboration 2025)
real(dp), parameter :: W0 = -0.752_dp, WA = -0.86_dp
! ACT DR6 (Calabrese et al. 2025, P-ACT-LB); Standard Model N_eff
real(dp), parameter :: NEFF_ACT = 2.86_dp, SIG_NEFF = 0.13_dp, NEFF_SM =
3.044_dp
! Galactic pole and node, J2000; obliquity of the ecliptic, J2000
real(dp), parameter :: A_NGP = 192.85948_dp, D_NGP = 27.12825_dp,
L_NCP = 122.93192_dp
real(dp), parameter :: EPS = 23.4392911_dp
integer, parameter :: MULT(5) = [6, 3, 3, 2, 1], Y6(5) = [1, -4, 2,
-3, 6]
logical, parameter :: TRIP(5) = [.true., .true., .true., .false.,
.false.]
logical, parameter :: DOUB(5) = [.true., .false., .false., .true.,
.false.]
integer, parameter :: C3(5) = [1, -1, -1, 0, 0]
integer, parameter :: PSY(3) = [3, -3, 0], PST(3) = [0, 0, 0], PSD(3) =
[1, 1, 0], PSK(3) = [0, 0, 1]
integer, parameter :: PFY(6) = [1, -4, 2, -3, 6, 0], PFT(6) = [1, -1,
-1, 0, 0, 0]
integer, parameter :: PFD(6) = [1, 0, 0, 1, 0, 0], PFK(6) = [0, 0, 0,
0, 0, 1]
integer, parameter :: CA(4) = [1, -1, 0, 0], CB(4) = [1, 3, 0, 0],
CK(4) = [0, 0, 1, 0], CD(4) = [0, 0, 0, 1]
integer :: nchk, nfail, y, t, k, d, ic, nfix, nbad, nrad, r, rc, ku,
v, nviol, nmis, i, nfixm
integer :: qq, nfree, nmmono, nport, j1, j2, j3, j4, jcj, jo, nstab,
nstabn, nmis2
integer :: gg, qv, bv, nv12, nv13, nd0, nd1, nv14, nv15, nv16, mm
logical :: okq
integer :: sq, sk, sb3, sl, sf, sg, ns1, ns2, ns3, ntot
logical :: fixed, okD0, okD, okBad
real(dp) :: omc, omb, omnu, q0, zacc, zeq, h0, rhoc, u, e4, lp, lr,
lew, w, zc, x, sep, smin, smax
real(dp) :: sd, alpha, delta, sb, beta, lam, s1, n_heavy, n_light,
vh(5), vl(5), rr, bh, bl, dn1, dn2
nchk = 0; nfail = 0
! ----- A. the fixed set and its readings, on the grid of
charges -6..6
nfix = 0; nbad = 0; nrad = 0
do y = -6, 6
do t = -6, 6
do k = -6, 6
do d = -6, 6
```

```
fixed = (-y == y) .and. (-t == t) .and. (-k == k) .and. (-d ==
d)
if (fixed) nfix = nfix + 1
do ic = 1, 4
r = CA(ic)*y + CB(ic)*t + CK(ic)*k + CD(ic)*d
rc = CA(ic)*(-y) + CB(ic)*(-t) + CK(ic)*(-k) + CD(ic)*(-d)
if (rc /= -r) nbad = nbad + 1
if (fixed .and. r /= 0) nbad = nbad + 1
end do
if ((y*y + t*t + k*k + d*d == 0) .neqv. fixed) nrad = nrad + 1
end do
end do
end do
call check('A1 the fixed set of conjugation on the 13^4 grid is one
point, the neutral state', nfix == 1)
call check('A2 photon, Z, gluon and dark-photon readings are odd and
vanish on the fixed set', nbad == 0)
call check('A3 the radiative weight vanishes exactly on the fixed
set', nrad == 0)
call check('A4 neutrino-like state: charge 0, radiative weight 18, Z
reading T3 - 0 sin2(thW) = 1/2', &
(3 + (-3) == 0) .and. (9 + 9 == 18) .and. &
(abs(3.0d0/6.0d0 - dble(3 + (-3))/6.0d0*0.23122d0 - 0.5d0)
< 1.0d-15))
nfixm = 0
do t = 6, -6, -6
if (-t == t) nfixm = nfixm + 1
end do
call check('A5 isospin triplet: one neutral member of three, so the
multiplet is not dark', nfixm == 1)
call closure([0, 0], 2, okD0); call closure([1, -1], 2, okD); call
closure([1, 0], 1, okBad)
call check('A6 the vector-like dark pair closes uncharged and charged
+1, -1; one chiral charge fails', &
okD0 .and. okD .and. (.not. okBad))
nmono = 0; nport = 0; nstab = 0; nstabn = 0
do j1 = 1, 3
do j2 = j1, 3
call scal([j1, j2])
do j3 = j2, 3
call scal([j1, j2, j3])
do j4 = j3, 3
call scal([j1, j2, j3, j4])
end do
end do
end do
do j1 = 1, 6
do j2 = j1, 6
do jcj = -1, 1, 2
do jo = 0, 3
call ferm(j1, j2, jcj, jo)
end do
end do
end do
end do
call check('A7 portal census: 199 renormalizable monomials, four
portals: S H+H, S2 H+H, L N H and its conjugate', &
nmono == 199 .and. nport == 4)
nfree = 0
do qq = -12, 12
call closure([qq, -qq], 2, okq)
if (okq) nfree = nfree + 1
end do
call check('A8 every vector-like dark charge q, -q closes all twelve
conditions, q = -12..12', nfree == 25)
call check('A9 a stable singlet keeps one door: under dark parity one
portal survives, S2 H+H, none for the fermion', &
nstab == 1 .and. nstabn == 0)
nmis2 = 0
do y = -6, 6
do t = -6, 6
do k = -1, 1
do d = -1, 1
if ((-y == y .and. -t == t .and. -k == k .and. -d == d) .neqv.
&
((-y == y .and. -t == t .and. -k == k) .and. d == 0))
nmis2 = nmis2 + 1
end do
end do
end do
```



```

end do
end do
call check('A10 C fixes a state exactly when C_SM fixes it and its
dark charge is zero, on the 13^2 x 3^2 grid', nmis2 == 0)
call check('A11 every coloured Standard Model field is charged in
every component: u 2/3, d -1/3, u^c -2/3, d^c 1/3', &
all([4, -2, -4, 2] /= 0))
nv12 = 0
do gg = 1, 5
do qv = -5, 5
if (gg*qv*qv < gg .and. qv /= 0) nv12 = nv12 + 1
end do
end do
do bv = 1, 5
do qv = -5, 5
if (.not. (1*qv*qv < bv*(qv*qv + 1))) nv12 = nv12 + 1
end do
end do
call check('A12 a bound fixes a charge only through a measured
coupling; a free coupling 1/(q^2+1) meets every bound', &
nv12 == 0)
nv13 = 0; nd0 = 0; nd1 = 0
do y = -6, 6
do t = -6, 6
do k = -1, 1
do d = -2, 2
if (t + y == 0 .and. 1000*t - 231*(t + y) == 0 .and. k == 0)
then
if (.not. (y == 0 .and. t == 0)) nv13 = nv13 + 1
if (d == 0) nd0 = nd0 + 1
if (d /= 0) nd1 = nd1 + 1
end if
end do
end do
end do
end do
call check('A13 the record sorts the title: every admitted state is
C_SM-fixed, admitted with and without dark charge', &
nv13 == 0 .and. nd0 > 0 .and. nd1 > 0)
nv14 = 0
do y = -6, 6
do d = -3, 3
if ((2*y == 0 .and. 2*d == 0) .neqv. (y == 0 .and. d == 0)) nv14 =
nv14 + 1
if (y == 0 .and. ((2*y == 0 .and. 2*d == 0) .neqv. (d == 0))) nv14
= nv14 + 1
end do
end do
call check('A14 a self-paired mass carries twice each abelian charge;
on the visible fixed set it is the dark bit', &
nv14 == 0)
nv15 = 0
do d = -3, 3
if (d /= 0 .and. -d == d) nv15 = nv15 + 1
end do
call check('A15 a dark-charged state takes its mass with a distinct
partner of opposite dark charge', nv15 == 0)
nv16 = 0
do d = -3, 3
do mm = 1, 3
if (1 + (-d) /= 2 - (1 + d)) nv16 = nv16 + 1
if ((1 + d == 1) .neqv. (d == 0)) nv16 = nv16 + 1
if (d /= 0 .and. (1 - d == 1 + d .or. 1 + d == 1)) nv16 = nv16 + 1
end do
end do
call check('A16 the carrier (1 + dark, mass) is equivariant onto the
fold; the line is the bit; a dark pair is off-line', &
nv16 == 0)
! ----- B. the measured budget
omc = OMC_H2/H100**2; omb = OMB_H2/H100**2; omnu =
SUM_MNU/93.14_dp/H100**2
call check('B1 Omega_c = 0.2645, Omega_b = 0.0493, ratio 5.36',
abs(omc - 0.2645_dp) < 5.0e-4_dp .and. &
abs(omb - 0.0493_dp) < 2.0e-4_dp .and. abs(omc/omb -
5.364_dp) < 0.01_dp)
call check('B2 cold dark matter, baryons and neutrinos reproduce
Omega_m to 0.002', &
abs(omc + omb + omnu - OM_M) < 0.002_dp)
q0 = 0.5_dp*OM_M - OM_L
call check('B3 q0 = -0.527 +/- 0.011: the expansion accelerates',
abs(q0 + 0.527_dp) < 1.0e-3_dp .and. &
q0/(1.5_dp*SIG_OM) < -40.0_dp)
zacc = (2.0_dp*OM_L/OM_M)**(1.0_dp/3.0_dp) - 1.0_dp; zeq =
(OM_L/OM_M)**(1.0_dp/3.0_dp) - 1.0_dp
call check('B4 acceleration begins at z = 0.631; vacuum and matter
densities cross at z = 0.295', &
abs(zacc - 0.631_dp) < 1.0e-3_dp .and. abs(zeq - 0.295_dp)
< 1.0e-3_dp)
h0 = H100*1.0e5_dp/MPC; rhoc = 3.0_dp*h0**2/(8.0_dp*PI*G)
call check('B5 critical density 8.52e-27 kg per cubic metre',
abs(rhoc/8.52e-27_dp - 1.0_dp) < 2.0e-3_dp)
u = OM_L*rhoc*C**2/EV*HBARC**3; e4 = u**0.25_dp
call check('B6 the vacuum energy density is (2.24 meV)^4',
abs(e4*1.0e3_dp - 2.24_dp) < 0.01_dp)
lp = log10(u/MPL**4); lr = log10(u/MPLRED**4); lew = log10(u/VEW**4)
call check('B7 rho_Lambda / M_P^4 = 10^-122.9, reduced 10^-120.1,
electroweak scale 10^-56.2', &
abs(lp + 122.95_dp) < 0.1_dp .and. abs(lr + 120.15_dp) <
0.1_dp .and. abs(lew + 56.16_dp) < 0.1_dp)
call check('B8 active density rho + 3p: dust +1, radiation +6, vacuum
-2', &
(1 + 3*0 == 1) .and. (3 + 3*1 == 6) .and. (1 + 3*(-1) ==
-2))
! ----- C. the kinetic floor
nviol = 0; nmis = 0
do ku = 0, 50
do v = -50, 50
if (ku + v <= 0) cycle
w = real(ku - v, dp)/real(ku + v, dp)
if (w < -1.0_dp) nviol = nviol + 1
if ((w == -1.0_dp) .neqv. (ku == 0)) nmis = nmis + 1
end do
end do
call check('C1 no canonical state (K >= 0, rho > 0) on the grid lies
below w = -1', nviol == 0)
call check('C2 w = -1 exactly when the kinetic term vanishes', nmis ==
0)
w = real(-1 - 10, dp)/real(-1 + 10, dp)
call check('C3 ghost control: K = -1, V = 10 gives w = -1.222, across
the divide', w < -1.0_dp)
x = (-1.0_dp - W0)/WA; zc = x/(1.0_dp - x)
call check('C4 the fitted CPL curve crosses w = -1 at z = 0.405 and
sits at -1.182 by z = 1', &
abs(zc - 0.405_dp) < 1.0e-3_dp .and. (W0 + WA*0.5_dp) <
-1.0_dp .and. W0 > -1.0_dp .and. W0 + WA < -1.0_dp)
! ----- D. the geometry of the fog
sd = sin(D_NGP*DEG)*sin(0.0_dp) +
cos(D_NGP*DEG)*cos(0.0_dp)*cos((L_NCP - 90.0_dp)*DEG)
delta = asin(sd)/DEG
alpha = A_NGP + atan2(sin((L_NCP - 90.0_dp)*DEG),
-sin(D_NGP*DEG)*cos((L_NCP - 90.0_dp)*DEG))/DEG
alpha = modulo(alpha, 360.0_dp)
call check('D1 the wind source (l = 90, b = 0) lies at alpha = 318.0,
delta = +48.3, in Cygnus', &
abs(alpha - 318.0_dp) < 0.05_dp .and. abs(delta - 48.33_dp)
< 0.02_dp)
sb = sin(delta*DEG)*cos(EPS*DEG) -
cos(delta*DEG)*sin(EPS*DEG)*sin(alpha*DEG)
beta = asin(sb)/DEG
lam = atan2(sin(alpha*DEG)*cos(EPS*DEG) + tan(delta*DEG)*sin(EPS*DEG),
cos(alpha*DEG))/DEG
smin = 180.0_dp; smax = 0.0_dp
do i = 0, 3599
s1 = real(i, dp)*0.1_dp
sep = acos(cos(beta*DEG)*cos((s1 - lam)*DEG))/DEG
smin = min(smin, sep); smax = max(smax, sep)
end do
call check('D2 the Sun never comes nearer the wind source than 59.6
degrees, nor farther than 120.4', &
abs(smin - 59.6_dp) < 0.1_dp .and. abs(smax - 120.4_dp) <
0.1_dp .and. abs(beta - smin) < 0.01_dp)
! ----- E. the floors of a fifth reading
bh = 1.25_dp*GEV_G/1.0e-24_dp; bl = 0.47_dp*GEV_G/1.0e-24_dp
call check('E1 self-interaction bounds: 1.25 cm2/g = 2.23 b/GeV
(Bullet), 0.47 cm2/g = 0.84 b/GeV (72 clusters)', &
abs(bh - 2.228_dp) < 0.005_dp .and. abs(bl - 0.838_dp) <
0.005_dp)
dn1 = (4.0_dp/7.0_dp)*1.0_dp*(10.75_dp/106.75_dp)**(4.0_dp/3.0_dp)
dn2 = (4.0_dp/7.0_dp)*2.0_dp*(10.75_dp/106.75_dp)**(4.0_dp/3.0_dp)
call check('E2 a relic once in equilibrium above the electroweak
scale: Delta N_eff >= 0.027 (scalar), 0.054 (dark photon)', &
abs(dn1 - 0.0268_dp) < 5.0e-4_dp .and. abs(dn2 - 0.0535_dp)
< 5.0e-4_dp)
call check('E3 the dark-photon floor lies inside the ACT DR6 95%
bound: still open', &
NEFF_SM + dn2 < NEFF_ACT + 2.0_dp*SIG_NEFF)
call check('E4 the Landauer price of one spent bit at 300 K is
2.87e-21 J', &
abs(KB*300.0_dp*log(2.0_dp)/2.871e-21_dp - 1.0_dp) <
1.0e-3_dp)
! ----- F. one halo, two species
n_heavy = 0.3_dp/100.0_dp; n_light = 0.3_dp/1.0e-14_dp
do i = 1, 5
rr = real(i, dp)*0.8_dp
vh(i) = sqrt(nfw_mass(rr)/rr); vl(i) = sqrt(nfw_mass(rr)/rr)
end do
call check('F1 at 0.3 GeV/cm3 a 100 GeV particle and a 10 micro-eV
field differ 1e16 in number, one rotation curve', &
abs(log10(n_light/n_heavy) - 16.0_dp) < 1.0e-9_dp .and.
all(vh == vl))
! ----- S. the dark row on the hardware seat
! S1: charge conjugation on the substrate's particles (charge in
thirds, colour, baryon number in
thirds, lepton number, frequency, rate) is carried onto the fold by
the charge carrier (1 + q3, freq).
ns1 = 0; ns2 = 0; ns3 = 0; ntot = 0
do sq = -3, 3
do sk = -1, 1
do sb3 = -3, 3
do sl = -1, 1
do sf = 0, 4
do sg = -2, 2
ntot = ntot + 1
if (1 + (-sq) /= 2 - (1 + sq)) ns1 = ns1 + 1

```



```

! S2: fixed by both involutions exactly when neutral and
stable, and then on the line twice:
! under the charge carrier (h = 1 + q3) and the energy
carrier (h = 1 + rate).
if ((sq == 0 .and. sk == 0 .and. sb3 == 0 .and. sl == 0)
.and. sg == 0) .neqv. &
  ((-sq == sq .and. -sk == sk .and. -sb3 == sb3 .and.
-sl == sl) .and. -sg == sg)) ns2 = ns2 + 1
if ((sq == 0 .and. sk == 0 .and. sb3 == 0 .and. sl == 0
.and. sg == 0) .and. &
  .not. (1 + sq == 1 .and. 1 + sg == 1)) ns2 = ns2 + 1
end do
end do
end do
end do
end do
call check('S1 on l1025 substrate particles the charge carrier sends
conjugation to the fold', ns1 == 0)
call check('S2 fixed by conjugation and the time mirror exactly when
neutral and stable; then on the line twice', ns2 == 0)
! S3: a dark-charged pair lands as two distinct off-line points,
exchanged by the fold, at one height,
! and the registration (|h - 1|, t) reads one record for both.
do d = -6, 6
  do mm = 1, 5
    if (d /= 0) then
      if (abs((1 + d) - 1) /= abs((1 - d) - 1)) ns3 = ns3 + 1
      if (1 + d == 1 - d .or. 1 + d == 1) ns3 = ns3 + 1
    else
      if (1 + d /= 1) ns3 = ns3 + 1
    end if
  end do
end do
call check('S3 every dark-charged pair is an off-line fold pair with
one registration; zero dark charge is the line', ns3 == 0)
write(*, '(a,3f9.4)') ' Omega_c, Omega_b, Omega_c/Omega_b      = ',
omc, omb, omc/omb
write(*, '(a,f9.4,a,f7.4)') ' q0                                = ',
q0, ' + - ', 1.5_dp*SIG_0M
write(*, '(a,2f9.4)') ' z acceleration onset, z Lambda = matter = ',
zacc, zeq
write(*, '(a,es11.4)') ' critical density (kg/m3)              = ',
rhoc
write(*, '(a,f9.4)') ' rho_Lambda^(1/4) (meV)                  = ',
e4*1.0e3_dp
write(*, '(a,3f9.2)') ' log10 rho_Lambda / (M_P, M_P red, v)^4 = ',
lp, lr, lew
write(*, '(a,f9.4)') ' CPL phantom crossing redshift          = ', zc
write(*, '(a,2f9.3)') ' wind source alpha, delta (deg)        = ',
alpha, delta
write(*, '(a,3f9.3)') ' ecliptic latitude, min and max Sun sep = ',
beta, smin, smax
write(*, '(a,2f9.4)') ' Delta N_eff floors, scalar and vector = ',
dn1, dn2
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', nchk,
',"failures":', nfail, '}'
if (nfail > 0) error stop 1
contains
logical function portal(ys, ts, ds, ks, nl)
! the kernel's portalM: hypercharge zero, triality zero mod 3,
doublets paired, a dark leg and a Standard Model leg
integer, intent(in) :: ys, ts, ds, ks, nl
portal = ys == 0 .and. modulo(ts, 3) == 0 .and. mod(ds, 2) == 0
.and. ks > 0 .and. ks < nl
end function portal
subroutine scal(ix)
integer, intent(in) :: ix(:)
integer :: j, ys, ts, ds, ks, ns
ys = 0; ts = 0; ds = 0; ks = 0; ns = 0
do j = 1, size(ix)
  ys = ys + PSY(ix(j)); ts = ts + PST(ix(j)); ds = ds + PSD(ix(j));
  ks = ks + PSK(ix(j))
end do
end subroutine

```

PASS A1 the fixed set of conjugation on the 13^4 grid is one point, the neutral state

PASS A2 photon, Z, gluon and dark-photon readings are odd and vanish on the fixed set

PASS A3 the radiative weight vanishes exactly on the fixed set

PASS A4 neutrino-like state: charge 0, radiative weight 18, Z reading $T3 - Q \sin^2(\theta_W) = 1/2$

PASS A5 isospin triplet: one neutral member of three, so the multiplet is not dark

PASS A6 the vector-like dark pair closes uncharged and charged +1, -1; one chiral charge fails

PASS A7 portal census: 199 renormalizable monomials, four portals: S H+H, S2 H+H, L N H and its conjugate

PASS A8 every vector-like dark charge q, -q closes all twelve conditions, q = -12.12

PASS A9 a stable singlet keeps one door: under dark parity one portal survives, S2 H+H, none for the fermion

PASS A10 C fixes a state exactly when C_{SM} fixes it and its dark charge is zero, on the $13^2 \times 3^2$ grid

PASS A11 every coloured Standard Model field is charged in every component: u 2/3, d -1/3, u^c -2/3, d^c 1/3

PASS A12 a bound fixes a charge only through a measured coupling; a free coupling $1/(q^2+1)$ meets every bound

```

if (ix(j) == 3) ns = ns + 1
end do
nmono = nmono + 1
if (portal(ys, ts, ds, ks, size(ix))) then
  nport = nport + 1
  if (mod(ns, 2) == 0) nstab = nstab + 1
end if
end subroutine scal
subroutine ferm(a, b, s, o)
integer, intent(in) :: a, b, s, o
integer :: ys, ts, ds, ks, nl, ns, nn
ys = s*(PFY(a) + PFY(b)); ts = s*(PFT(a) + PFT(b)); ds = PFD(a) +
PFD(b); ks = PFK(a) + PFK(b); nl = 2; ns = 0
nn = merge(1, 0, a == 6) + merge(1, 0, b == 6)
if (o > 0) then
  ys = ys + PSY(o); ts = ts + PST(o); ds = ds + PSD(o); ks = ks +
PSK(o); nl = 3
  if (o == 3) ns = 1
end if
end if
nmono = nmono + 1
if (portal(ys, ts, ds, ks, nl)) then
  nport = nport + 1
  if (mod(ns, 2) == 0 .and. mod(nn, 2) == 0) then
    nstab = nstab + 1
    if (nn > 0) nstabn = nstabn + 1
  end if
end if
end subroutine ferm
subroutine closure(q, n, ok)
! the kernel's closes, sum for sum: six visible conditions and six
dark ones over the Standard Model and n singlets
integer, intent(in) :: q(:), n
logical, intent(out) :: ok
integer :: i, m, sm(12), fm(7), fy(7), fc(7), fq(7)
logical :: ft(7), fdb(7)
m = 5 + n
fm(1:5) = MULT; fy(1:5) = Y6; ft(1:5) = TRIP; fdb(1:5) = DOUB;
fc(1:5) = C3; fq(1:5) = 0
do i = 1, n
  fm(5+i) = 1; fy(5+i) = 0; ft(5+i) = .false.; fdb(5+i) = .false.;
  fc(5+i) = 0; fq(5+i) = q(i)
end do
sm = 0
do i = 1, m
  sm(1) = sm(1) + fm(i)*fy(i)
  sm(2) = sm(2) + fm(i)*fy(i)**3
  if (ft(i)) sm(3) = sm(3) + (fm(i)/3)*fy(i)
  if (fdb(i)) sm(4) = sm(4) + (fm(i)/2)*fy(i)
  if (fdb(i)) sm(5) = sm(5) + fm(i)/2
  if (ft(i)) sm(6) = sm(6) + (fm(i)/3)*fc(i)
  sm(7) = sm(7) + fm(i)*fq(i)
  sm(8) = sm(8) + fm(i)*fq(i)**3
  sm(9) = sm(9) + fm(i)*fq(i)**2*fy(i)
  sm(10) = sm(10) + fm(i)*fq(i)*fy(i)**2
  if (ft(i)) sm(11) = sm(11) + (fm(i)/3)*fq(i)
  if (fdb(i)) sm(12) = sm(12) + (fm(i)/2)*fq(i)
end do
ok = all(sm([1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12]) == 0) .and.
mod(sm(5), 2) == 0
end subroutine closure
pure function nfw_mass(rr) result(m)
real(dp), intent(in) :: rr
real(dp) :: m
m = 4.0_dp*PI*(log(1.0_dp + rr) - rr/(1.0_dp + rr))
end function nfw_mass
subroutine check(label, cond)
character(*), intent(in) :: label
logical, intent(in) :: cond
nchk = nchk + 1
if (.not. cond) nfail = nfail + 1
write(*, '(a,a)') merge(' PASS ', ' FAIL ', cond), label
end subroutine check
end program dark_twin

```

PASS A13 the record sorts the title: every admitted state is C_{SM} -fixed, admitted with and without dark charge

PASS A14 a self-paired mass carries twice each abelian charge; on the visible fixed set it is the dark bit

PASS A15 a dark-charged state takes its mass with a distinct partner of opposite dark charge

PASS A16 the carrier (1 + dark, mass) is equivariant onto the fold; the line is the bit; a dark pair is off-line

PASS B1 Ω_{c} = 0.2645, Ω_{b} = 0.0493, ratio 5.36

PASS B2 cold dark matter, baryons and neutrinos reproduce Ω_{m} to 0.002

PASS B3 $q_0 = -0.527 \pm 0.011$: the expansion accelerates

PASS B4 acceleration begins at $z = 0.631$; vacuum and matter densities cross at $z = 0.295$

PASS B5 critical density $8.52e-27$ kg per cubic metre

PASS B6 the vacuum energy density is $(2.24 \text{ MeV})^4$

PASS B7 $\rho_{\text{Lambda}} / M_P^4 = 10^{-122.9}$, reduced $10^{-120.1}$, electroweak scale $10^{-56.2}$

PASS B8 active density $\rho + 3p$: dust +1, radiation +6, vacuum -2

PASS C1 no canonical state ($K \geq 0$, $\rho > 0$) on the grid lies below $w = -1$

PASS C2 $w = -1$ exactly when the kinetic term vanishes

PASS C3 ghost control: $K = -1$, $V = 10$ gives $w = -1.222$, across the divide
 PASS C4 the fitted CPL curve crosses $w = -1$ at $z = 0.405$ and sits at -1.182 by $z = 1$
 PASS D1 the wind source ($l = 90$, $b = 0$) lies at $\alpha = 318.0$, $\delta = +48.3$, in Cygnus
 PASS D2 the Sun never comes nearer the wind source than 59.6 degrees, nor farther than 120.4
 PASS E1 self-interaction bounds: $1.25 \text{ cm}^2/\text{g} = 2.23 \text{ b/GeV}$ (Bullet), $0.47 \text{ cm}^2/\text{g} = 0.84 \text{ b/GeV}$ (72 clusters)
 PASS E2 a relic once in equilibrium above the electroweak scale: $\Delta N_{\text{eff}} \geq 0.027$ (scalar), 0.054 (dark photon)
 PASS E3 the dark-photon floor lies inside the ACT DR6 95% bound: still open
 PASS E4 the Landauer price of one spent bit at 300 K is $2.87\text{e-}21 \text{ J}$
 PASS F1 at 0.3 GeV/cm^3 a 100 GeV particle and a 10 micro-eV field differ $1\text{e}16$ in number, one rotation curve

PASS S1 on 11025 substrate particles the charge carrier sends conjugation to the fold
 PASS S2 fixed by conjugation and the time mirror exactly when neutral and stable; then on the line twice
 PASS S3 every dark-charged pair is an off-line fold pair with one registration; zero dark charge is the line
 $\Omega_c, \Omega_b, \Omega_c/\Omega_b = 0.2645 \quad 0.0493 \quad 5.3643$
 $q0 = -0.5271 \pm 0.0109$
 $z \text{ acceleration onset, } z_{\text{Lambda}} = \text{matter} = 0.6316 \quad 0.2950$
 $\text{critical density (kg/m}^3) = 8.5227\text{E-}27$
 $\rho_{\text{Lambda}}^{1/4} \text{ (MeV)} = 2.2395$
 $\log_{10} \rho_{\text{Lambda}} / (M_P, M_P \text{ red, } v)^4 = -122.95 \quad -120.15 \quad -56.16$
 $\text{CPL phantom crossing redshift} = 0.4052$
 $\text{wind source } \alpha, \delta \text{ (deg)} = 318.004 \quad 48.330$
 $\text{ecliptic latitude, min and max Sun sep} = 59.574 \quad 59.574 \quad 120.426$
 $\Delta N_{\text{eff}} \text{ floors, scalar and vector} = 0.0268 \quad 0.0535$
 BATTERY-JSON: {"checks":38,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

PhysOS^T 1.0.5p · RECEIPT · mode quick
 TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
 MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
 GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held by review, clean
 EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
 REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
 WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
 CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
 ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
 SEAT EARNED · this run

 SCREEN forbidden: none | imports: 0 | axioms declared: 0
 laws: 125
 JUDGMENT refused 125 of 125 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True